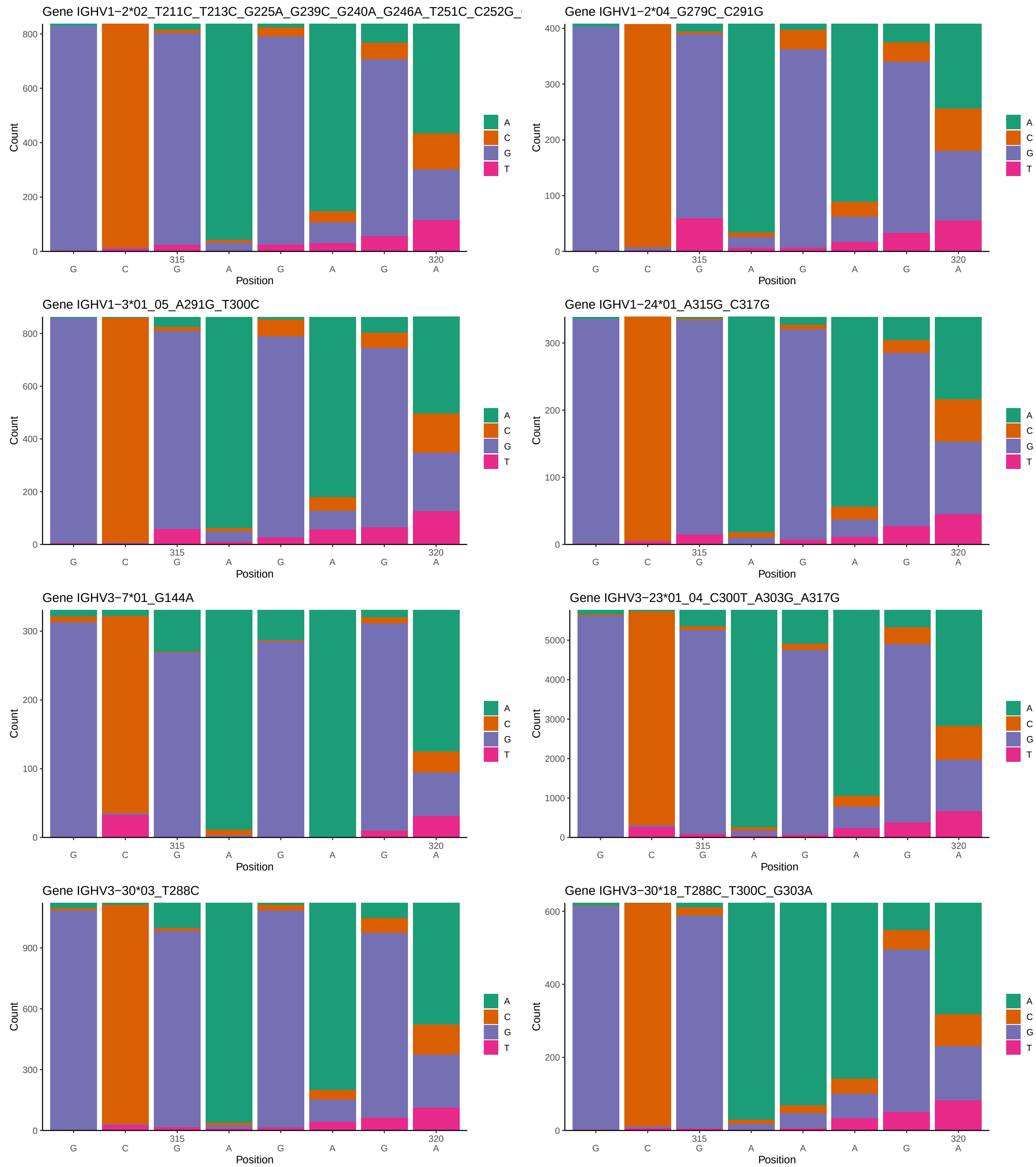
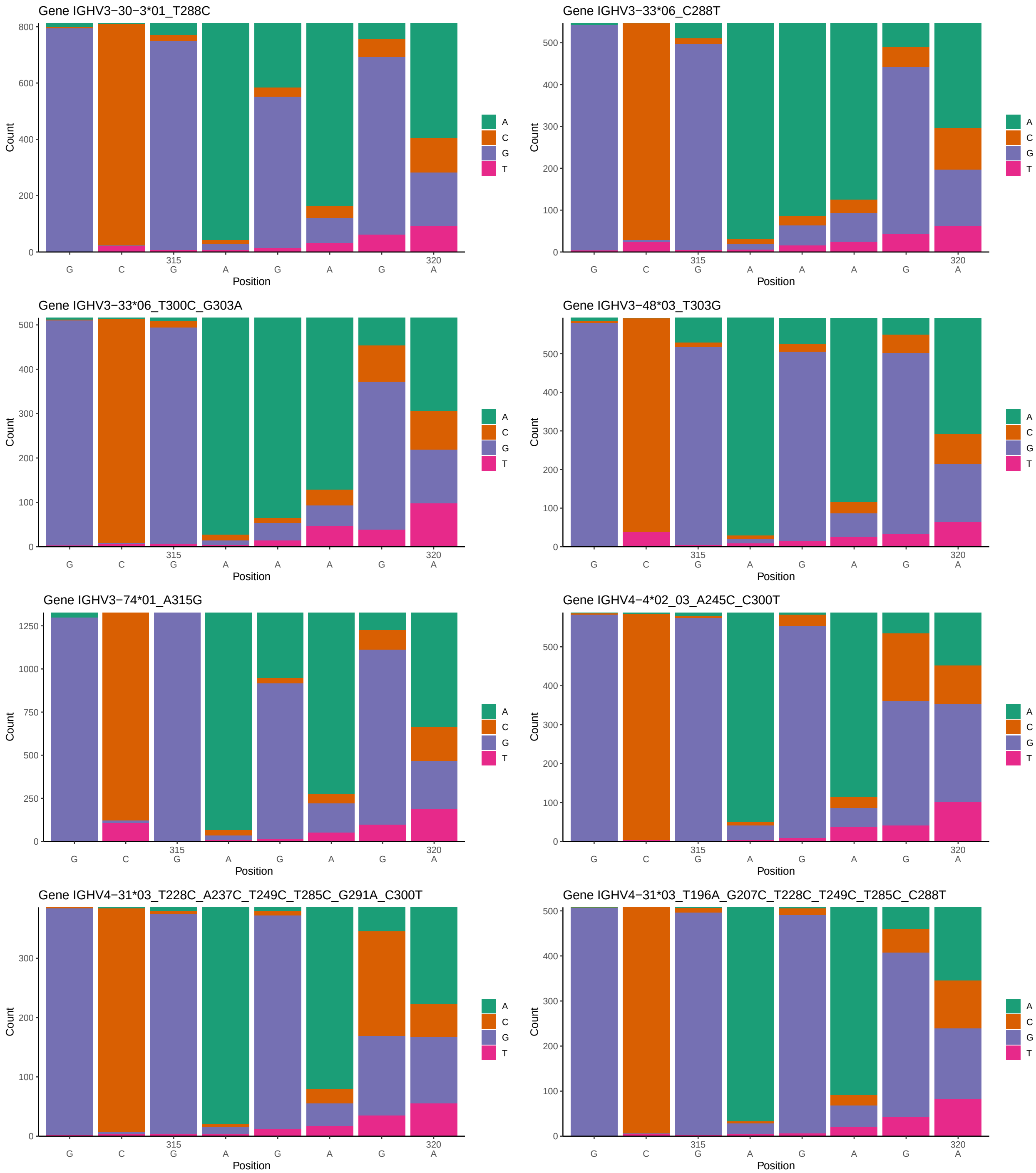
OGRDBstats Report

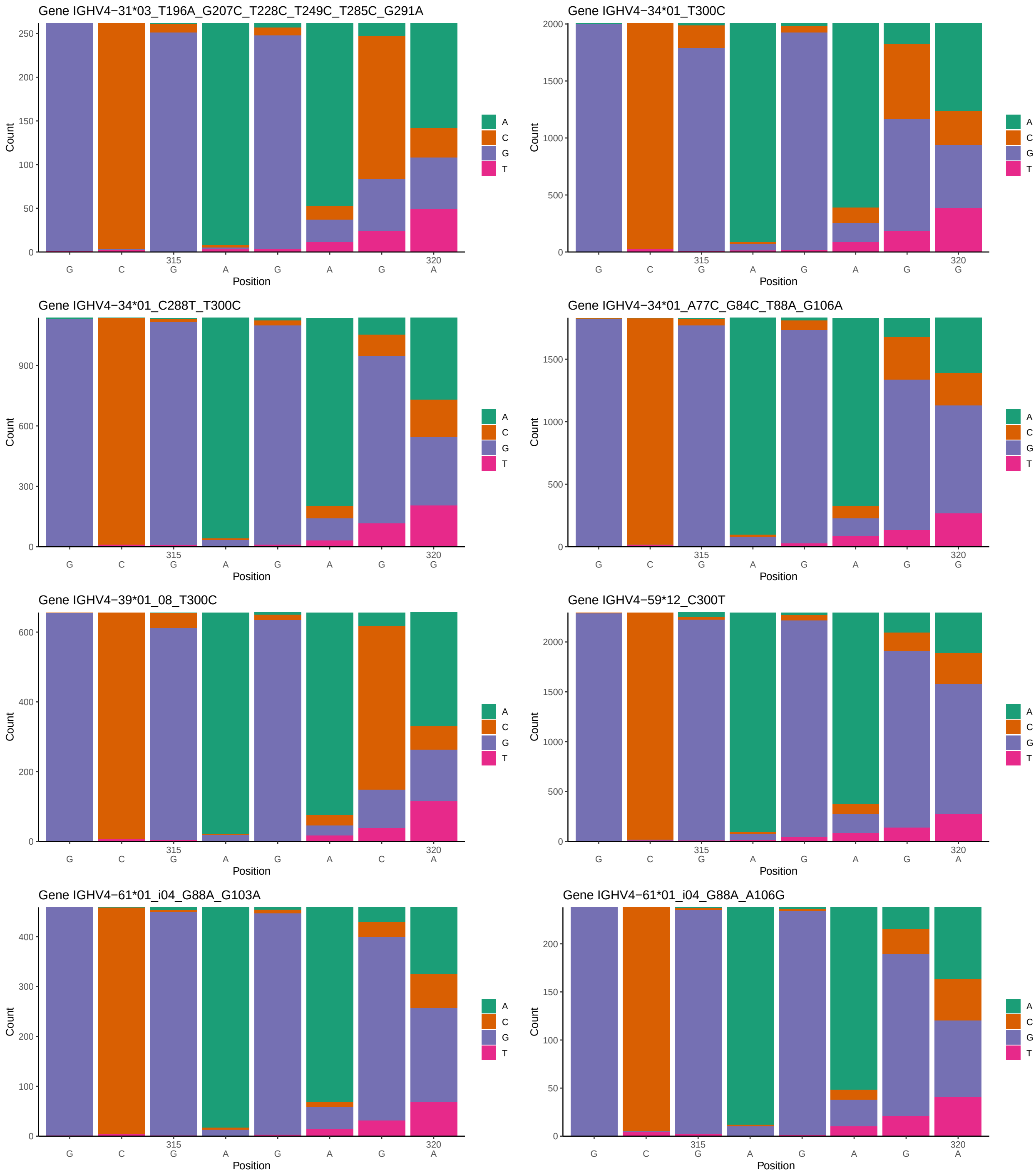
Contents

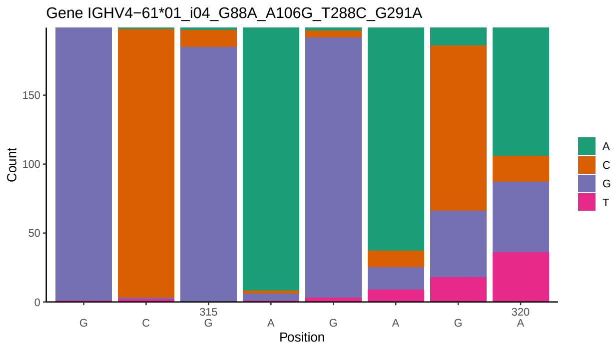
1 Novel sequence analysis

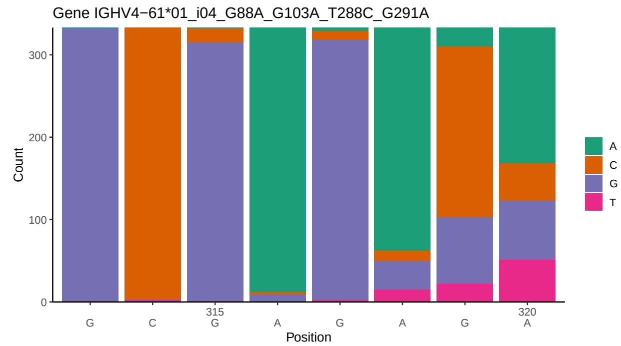
1.1 End-nucleotide composition

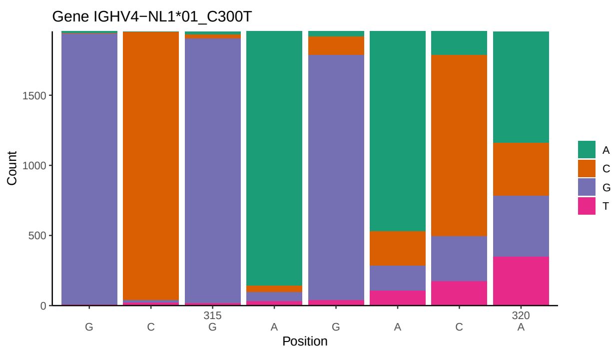


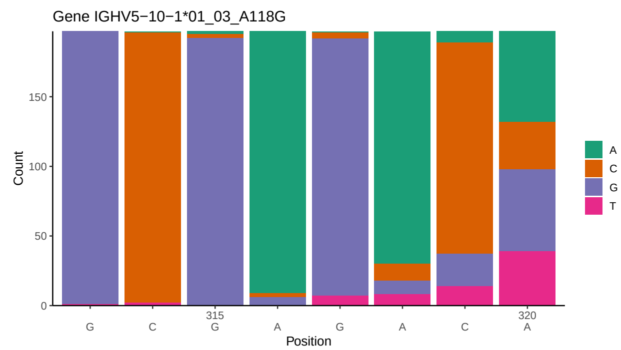

Gene IGHV3−30−3*01_T288C
Gene IGHV3−33*06_C288T
Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−48*03_T303G
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−31*03_T228C_A237C_T249C_T285C_G291A_C300T
Gene IGHV4−31*03_T196A_G207C_T228C_T249C_T285C_C288T
Count
Position
A
C
G
T


Gene IGHV4−31*03_T196A_G207C_T228C_T249C_T285C_G291A
Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_C288T_T300C
Gene IGHV4−34*01_A77C_G84C_T88A_G106A
Gene IGHV4−39*01_08_T300C
Gene IGHV4−59*12_C300T
Gene IGHV4−61*01_i04_G88A_G103A
Gene IGHV4−61*01_i04_G88A_A106G
Count
Position
A
C
G
T

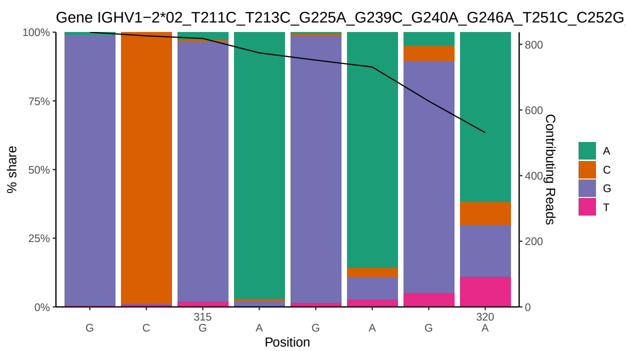


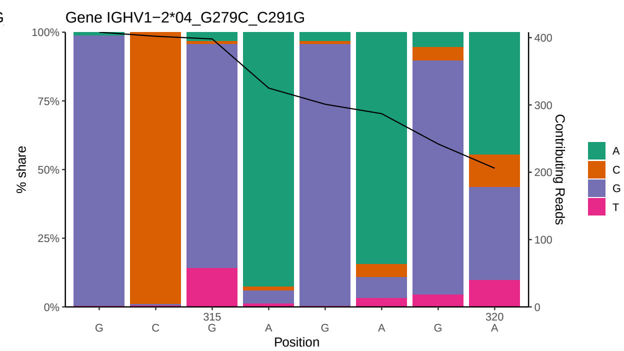


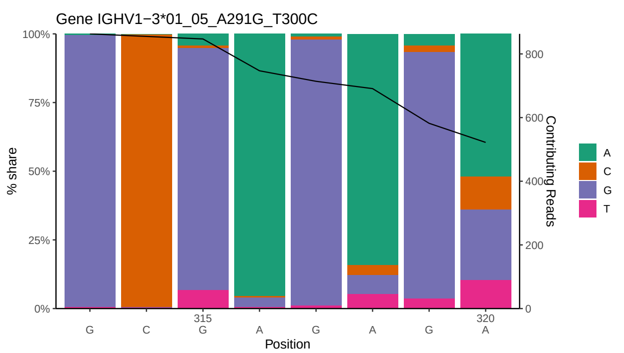


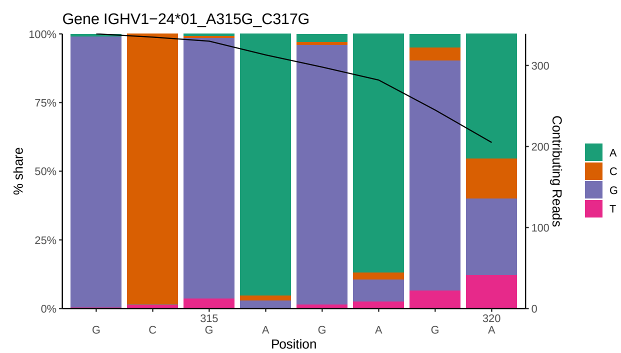


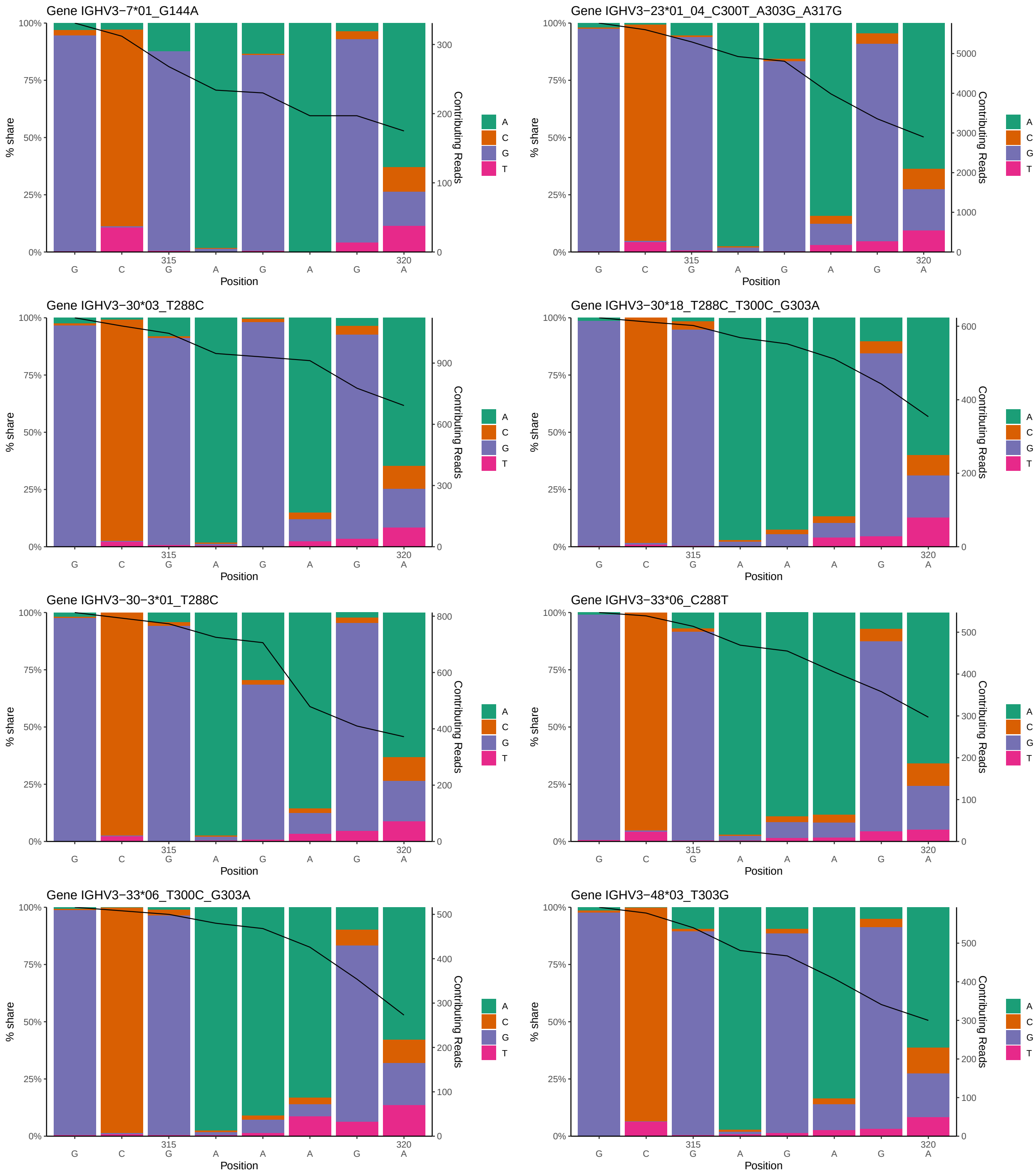
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

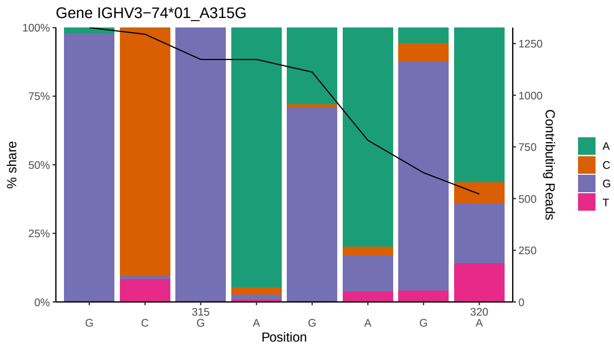









Gene IGHV3−7*01_G144A
Gene IGHV3−23*01_04_C300T_A303G_A317G
Gene IGHV3−30*03_T288C
Gene IGHV3−30*18_T288C_T300C_G303A
Gene IGHV3−30−3*01_T288C
Gene IGHV3−33*06_C288T
Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−48*03_T303G
% share
Contributing Reads
Position
G C G A G A G A
315
320
A
C
G
T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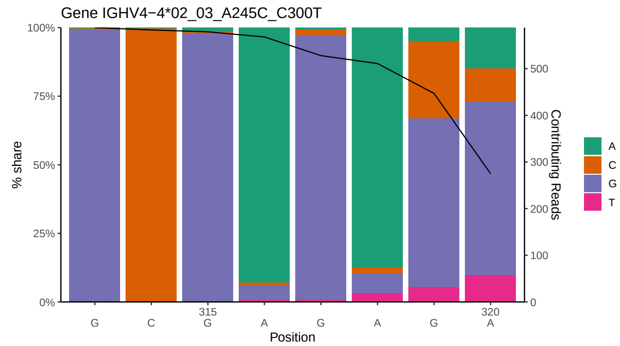
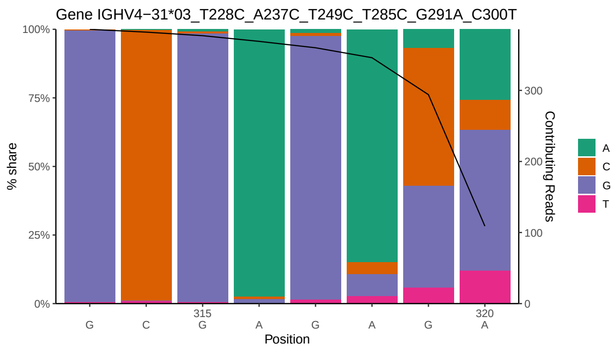
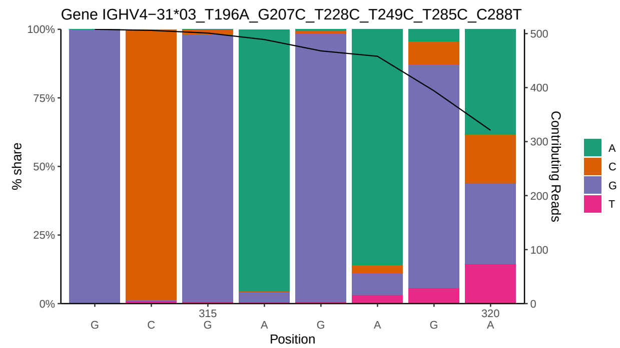
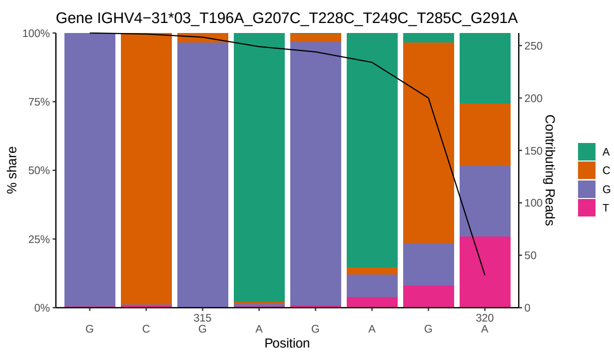
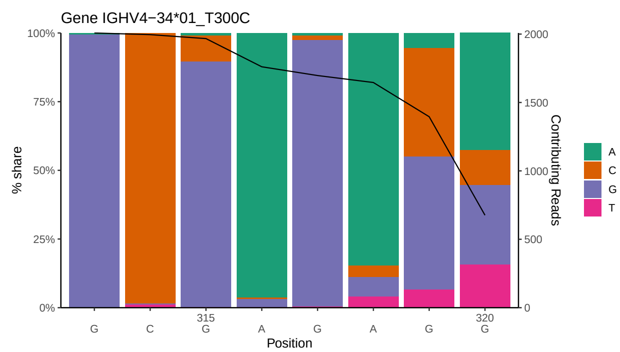
% share
100%
75%
50%
25%
0%
1250
1000
750
500
250
0
Contributing Reads
G
C
315
G
A
G
A
G
320
A
Position
A
C
G
T

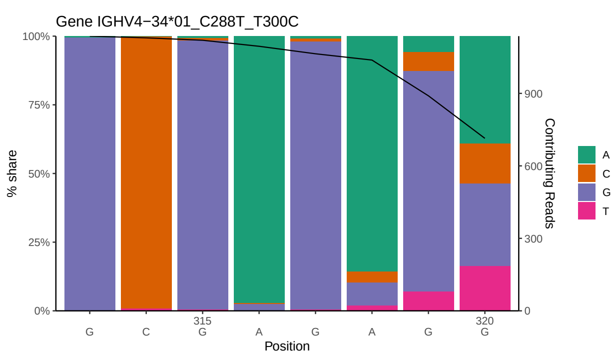

Gene IGHV4−4*02_03_A245C_C300T
% share
100%
75%
50%
25%
0%
500
400
300
200
100
0
Contributing Reads
G
C
315
G
A
G
A
G
320
A
Position
A
C
G
T

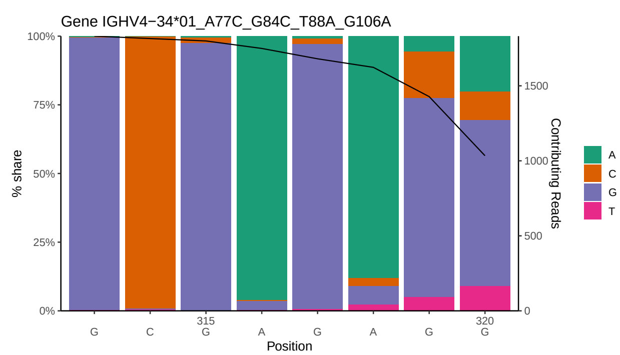

Gene IGHV4−31*03_T228C_A237C_T249C_T285C_G291A_C300T
% share
100%
75%
50%
25%
0%
300
200
100
0
Contributing Reads
G
C
315
G
A
G
A
G
320
A
Position
A
C
G
T

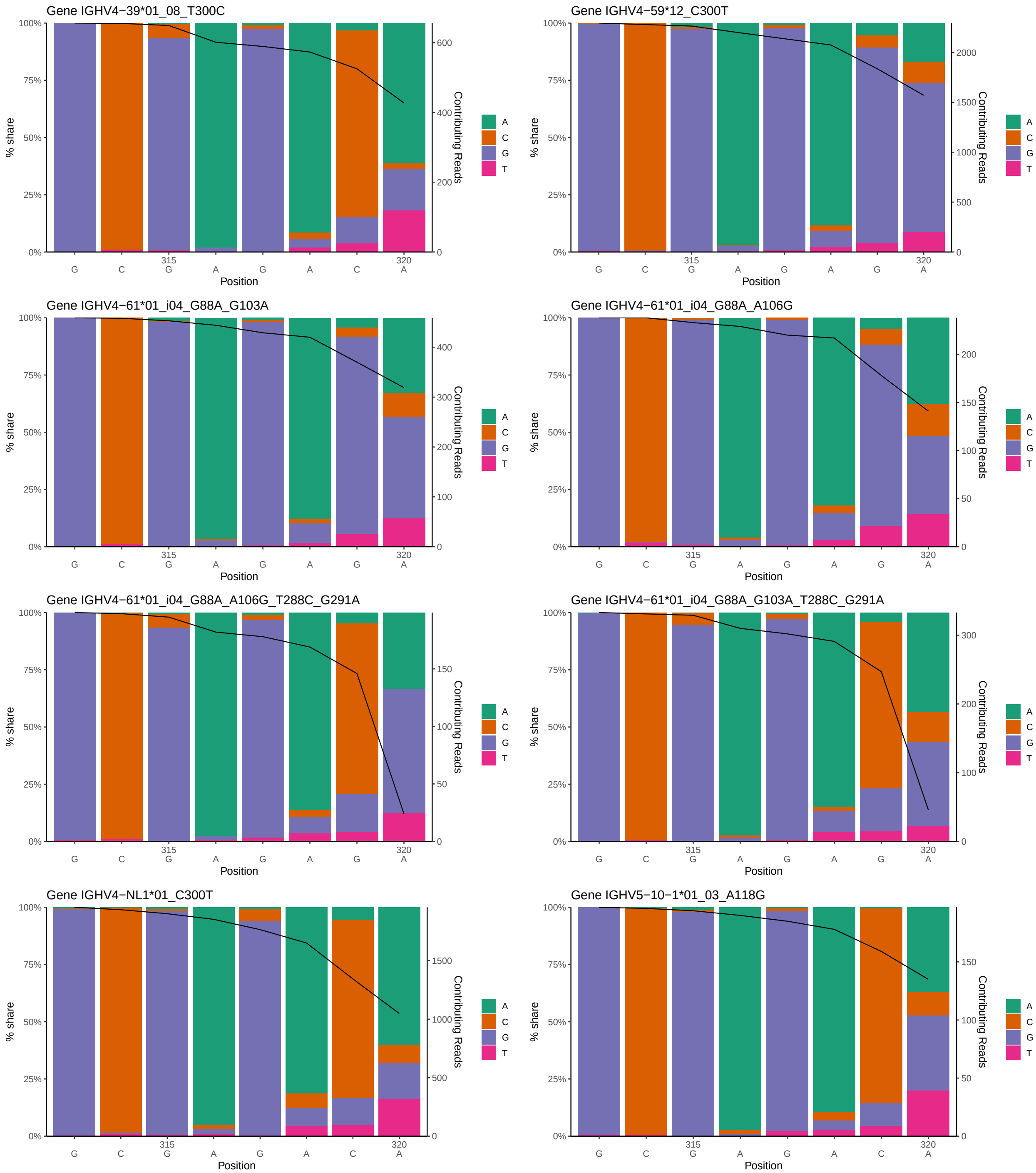

Gene IGHV4−31*03_T196A_G207C_T228C_T249C_T285C_C288T
% share
100%
75%
50%
25%
0%
500
400
300
200
100
0
Contributing Reads
G
C
315
G
A
G
A
G
320
A
Position
A
C
G
T


Gene IGHV4−31*03_T196A_G207C_T228C_T249C_T285C_G291A
% share
100%
75%
50%
25%
0%
250
200
150
100
50
0
Contributing Reads
G
C
315
G
A
G
A
G
320
A
Position
A
C
G
T

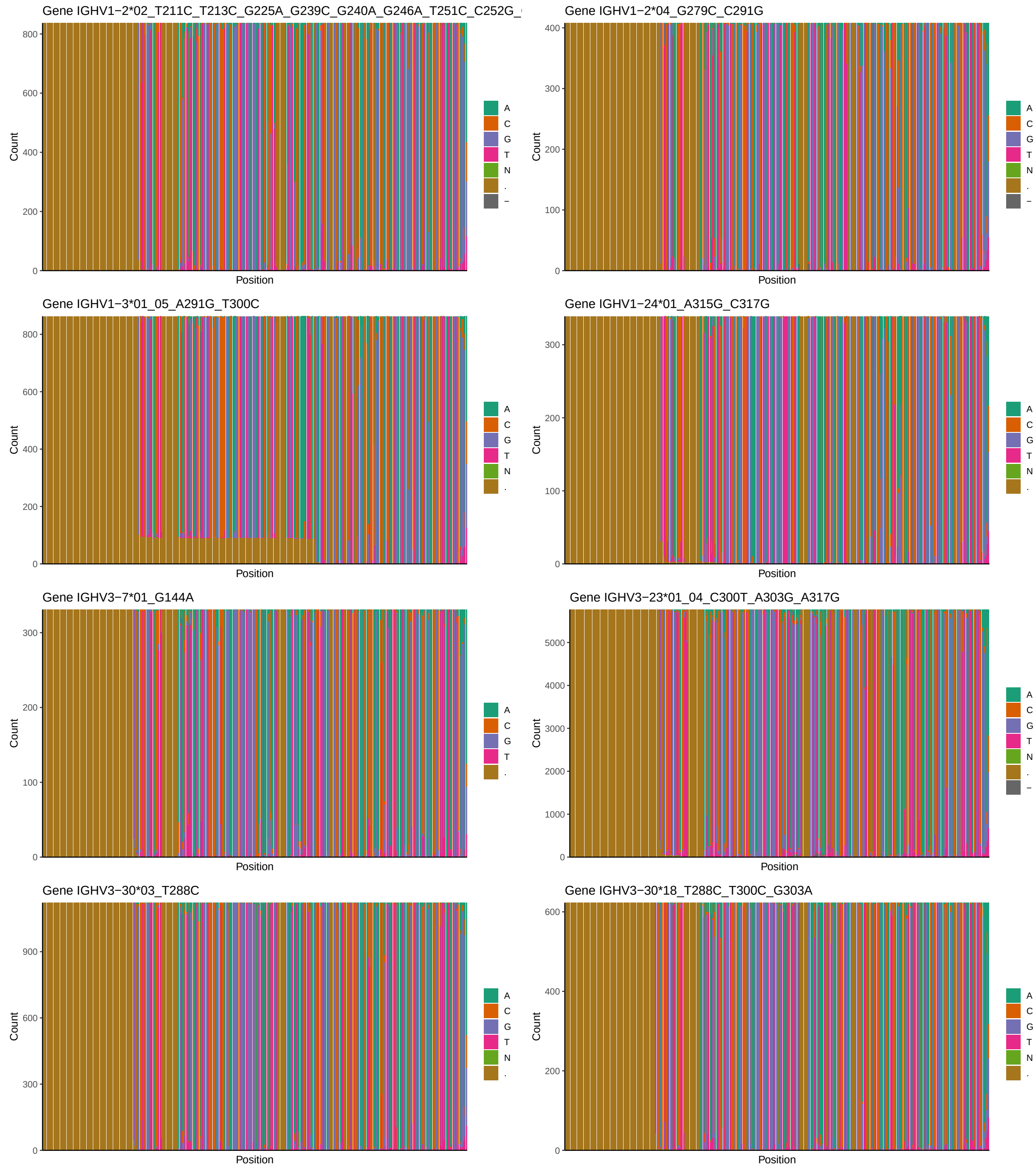

Gene IGHV4−34*01_T300C
% share
100%
75%
50%
25%
0%
2000
1500
1000
500
0
Contributing Reads
G
C
315
G
A
G
A
G
320
G
Position
A
C
G
T

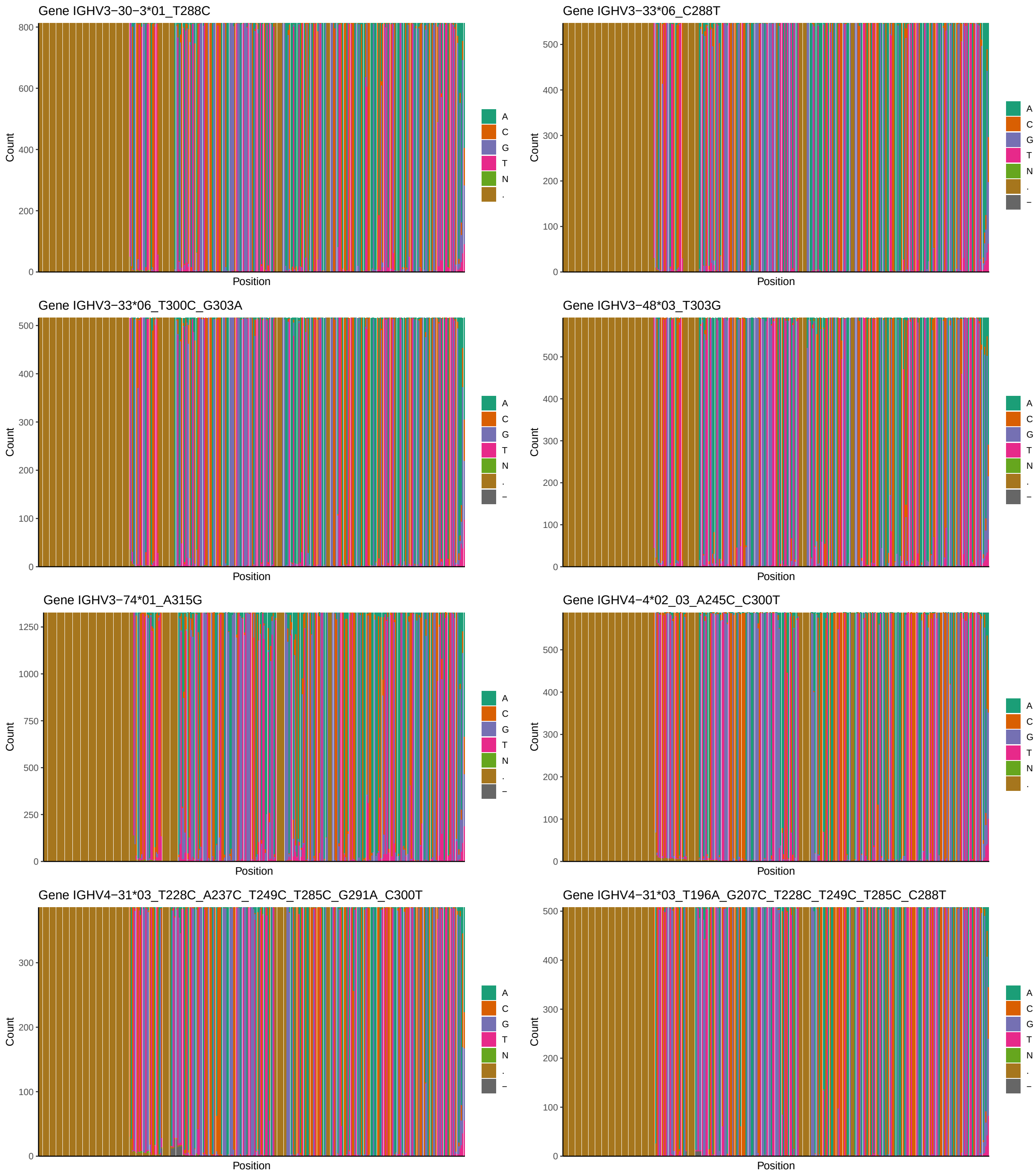

Gene IGHV4−34*01_C288T_T300C
% share
100%
75%
50%
25%
0%
900
600
300
0
Contributing Reads
G
C
315
G
A
G
A
G
320
G
Position
A
C
G
T

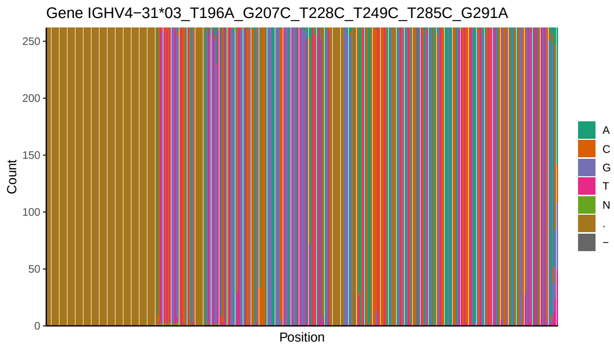

Gene IGHV4−34*01_A77C_G84C_T88A_G106A
% share
100%
75%
50%
25%
0%
1500
1000
500
0
Contributing Reads
G
C
315
G
A
G
A
G
320
G
Position
A
C
G
T

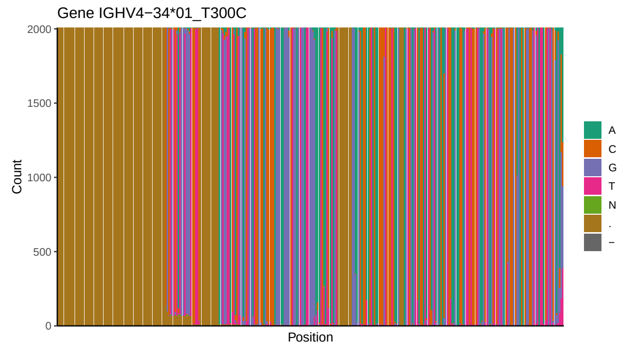

Gene IGHV4−39*01_08_T300C
Gene IGHV4−59*12_C300T
Gene IGHV4−61*01_i04_G88A_G103A
Gene IGHV4−61*01_i04_G88A_A106G
Gene IGHV4−61*01_i04_G88A_A106G_T288C_G291A
Gene IGHV4−61*01_i04_G88A_G103A_T288C_G291A
Gene IGHV4−NL1*01_C300T
Gene IGHV5−10−1*01_03_A118G
% share
Contributing Reads
Position
G C G A G A C A
315
320
A
C
G
T

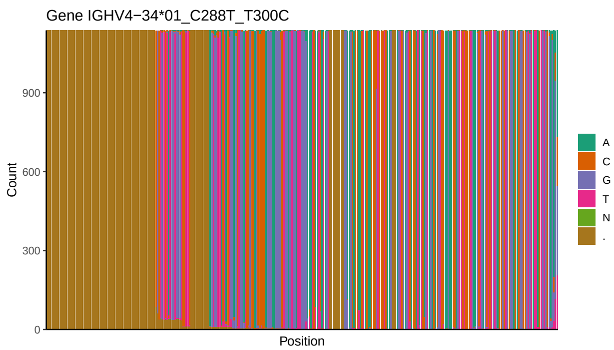
1.3 Whole-sequence composition of each assigned read

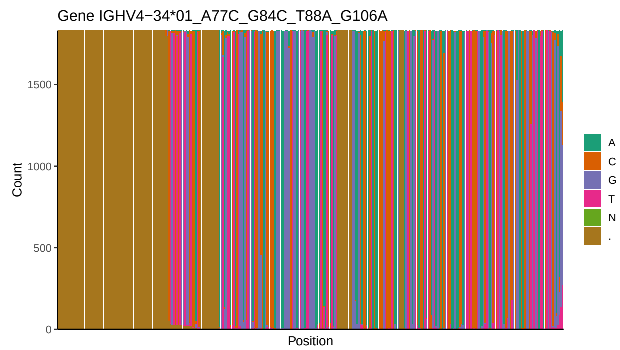


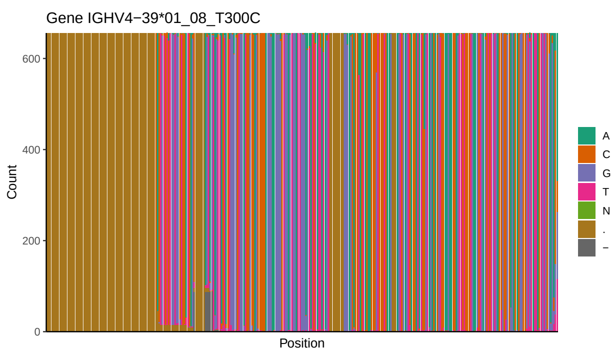

Gene IGHV3−30−3*01_T288C
Gene IGHV3−33*06_C288T
Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−48*03_T303G
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−31*03_T228C_A237C_T249C_T285C_G291A_C300T
Gene IGHV4−31*03_T196A_G207C_T228C_T249C_T285C_C288T
Count
Position
A
C
G
T
N
.
−

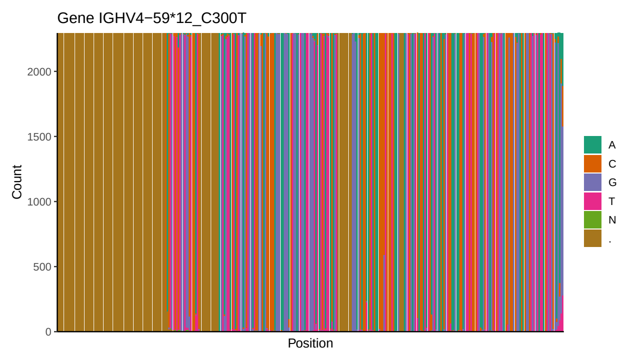

Gene IGHV4−31*03_T196A_G207C_T228C_T249C_T285C_G291A
Count
250
200
150
100
50
0
Position
A
C
G
T
N
.
−

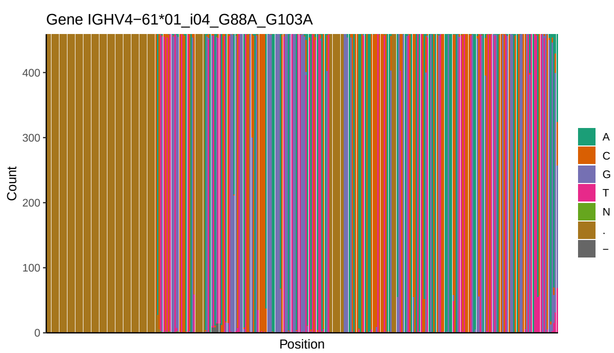

Gene IGHV4−34*01_T300C
Count
2000
1500
1000
500
0
Position
A
C
G
T
N
.
−

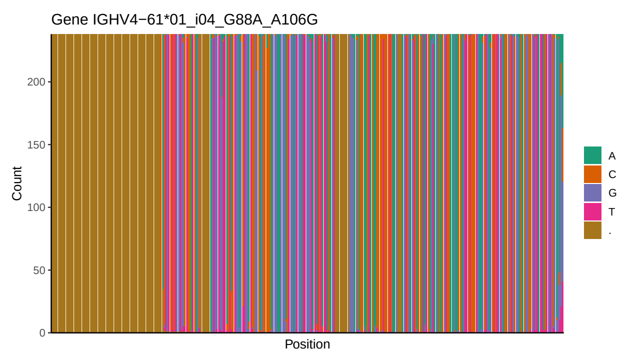

Gene IGHV4−34*01_C288T_T300C
Count
900
600
300
0
Position
A
C
G
T
N
.

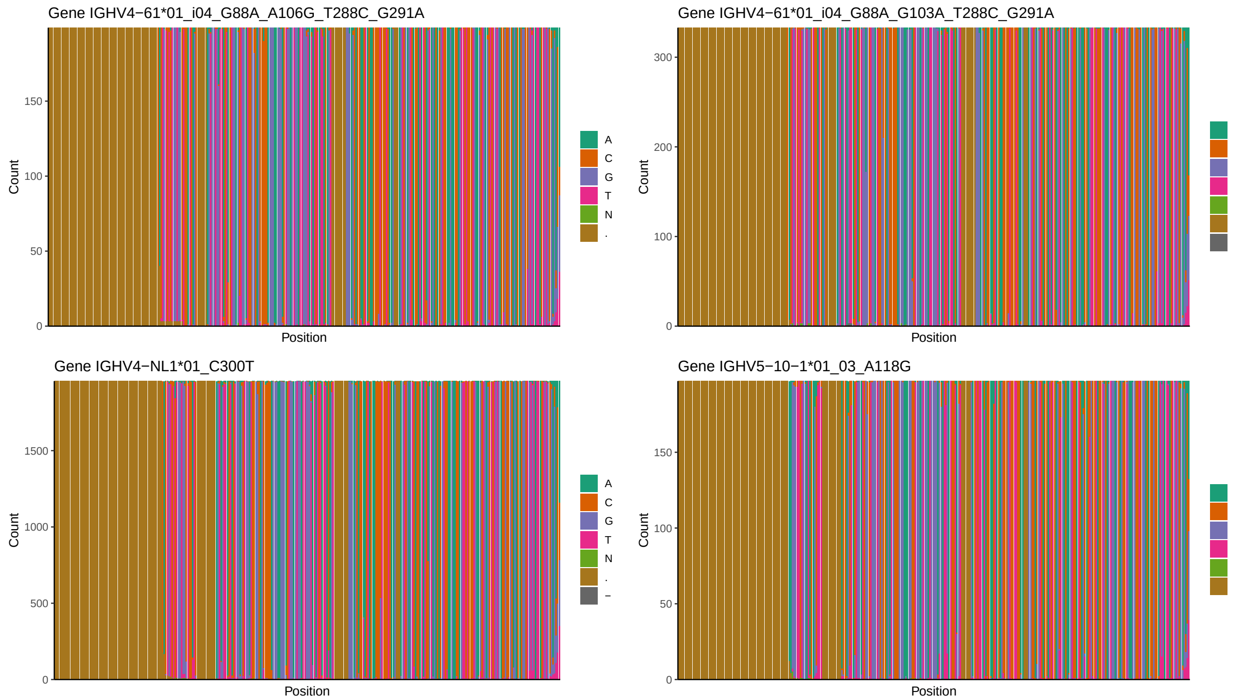

Gene IGHV4−34*01_A77C_G84C_T88A_G106A
Count
1500
1000
500
0
Position
A
C
G
T
N
.


Gene IGHV4−39*01_08_T300C
Count
600
400
200
0
Position
A
C
G
T
N
.
−

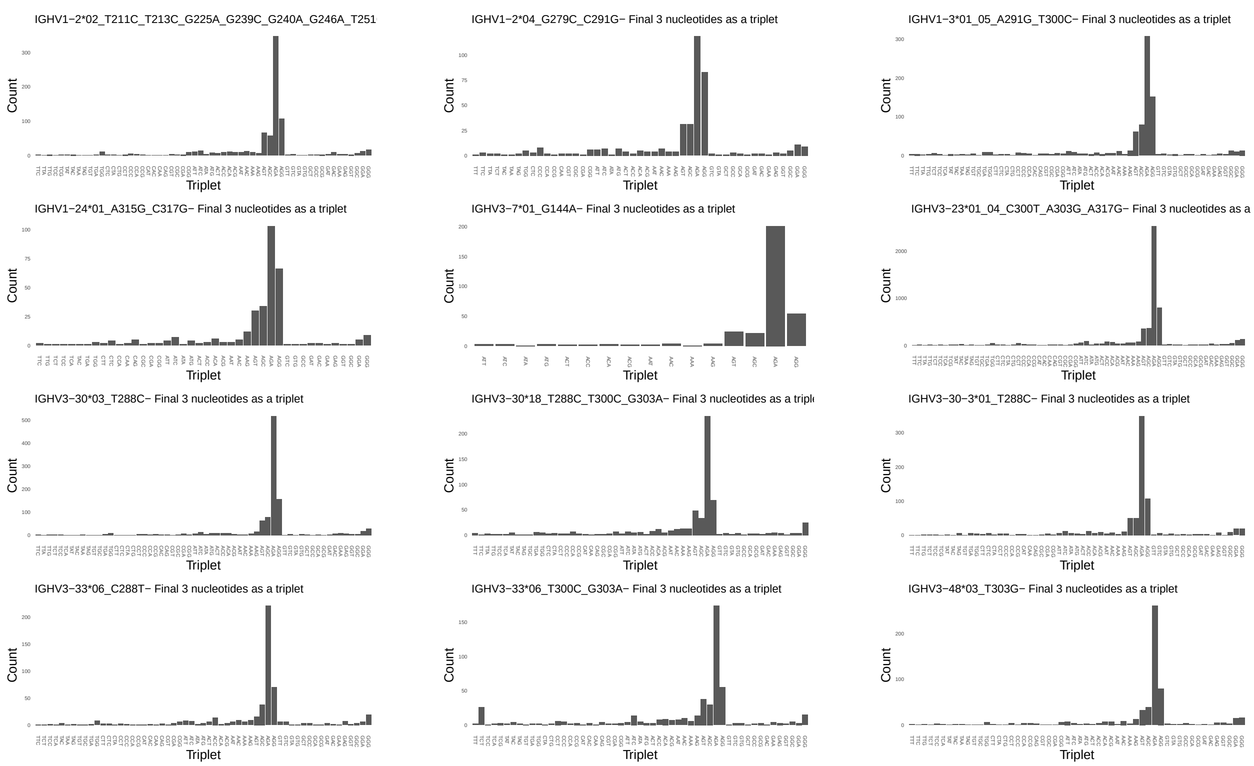

Gene IGHV4−59*12_C300T
Count
2000
1500
1000
500
0
Position
A
C
G
T
N
.

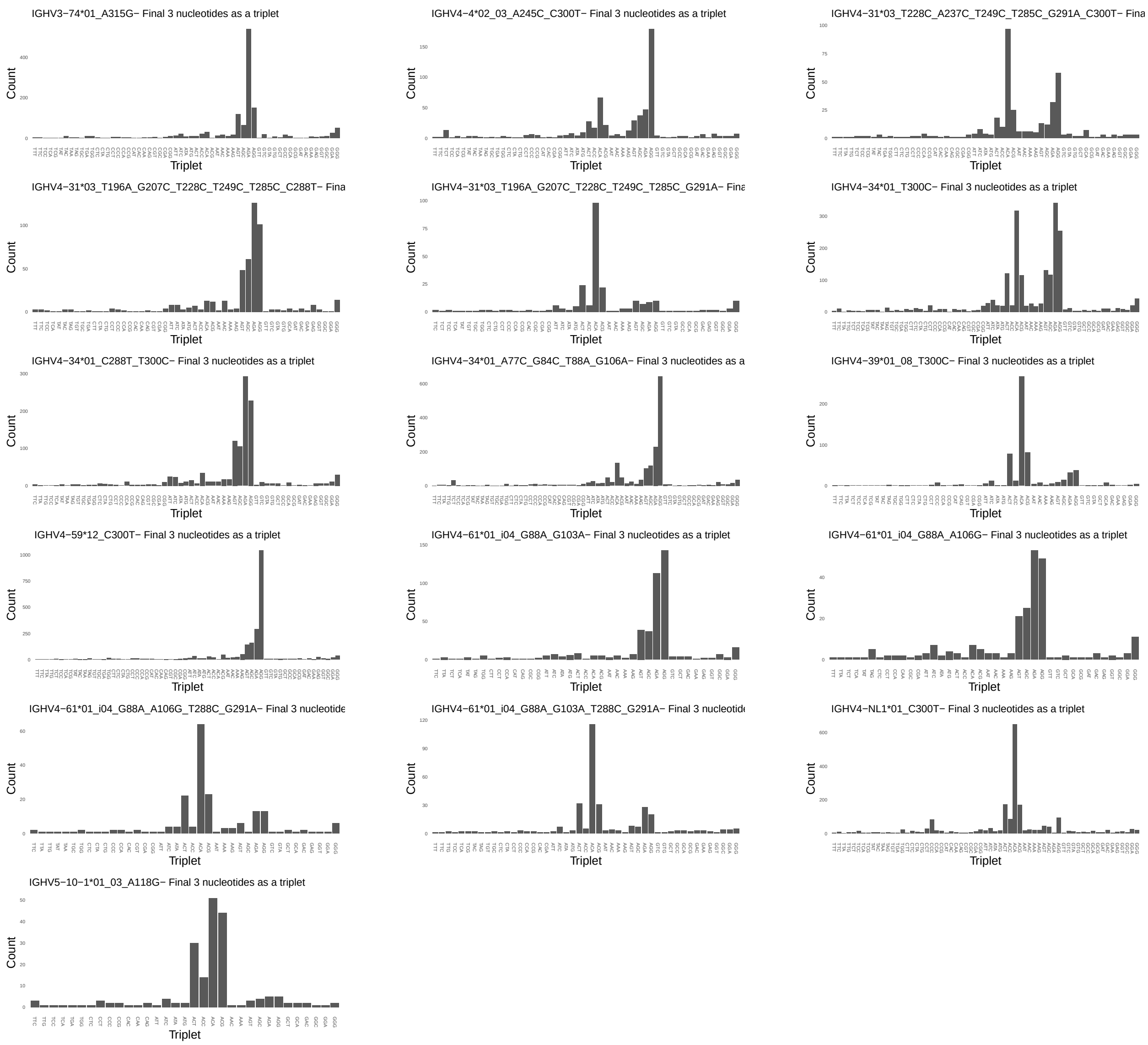

Gene IGHV4−61*01_i04_G88A_G103A
Count
400
300
200
100
0
Position
A
C
G
T
N
.
−


Gene IGHV4−61*01_i04_G88A_A106G
Count
200
150
100
50
0
Position
A
C
G
T
.

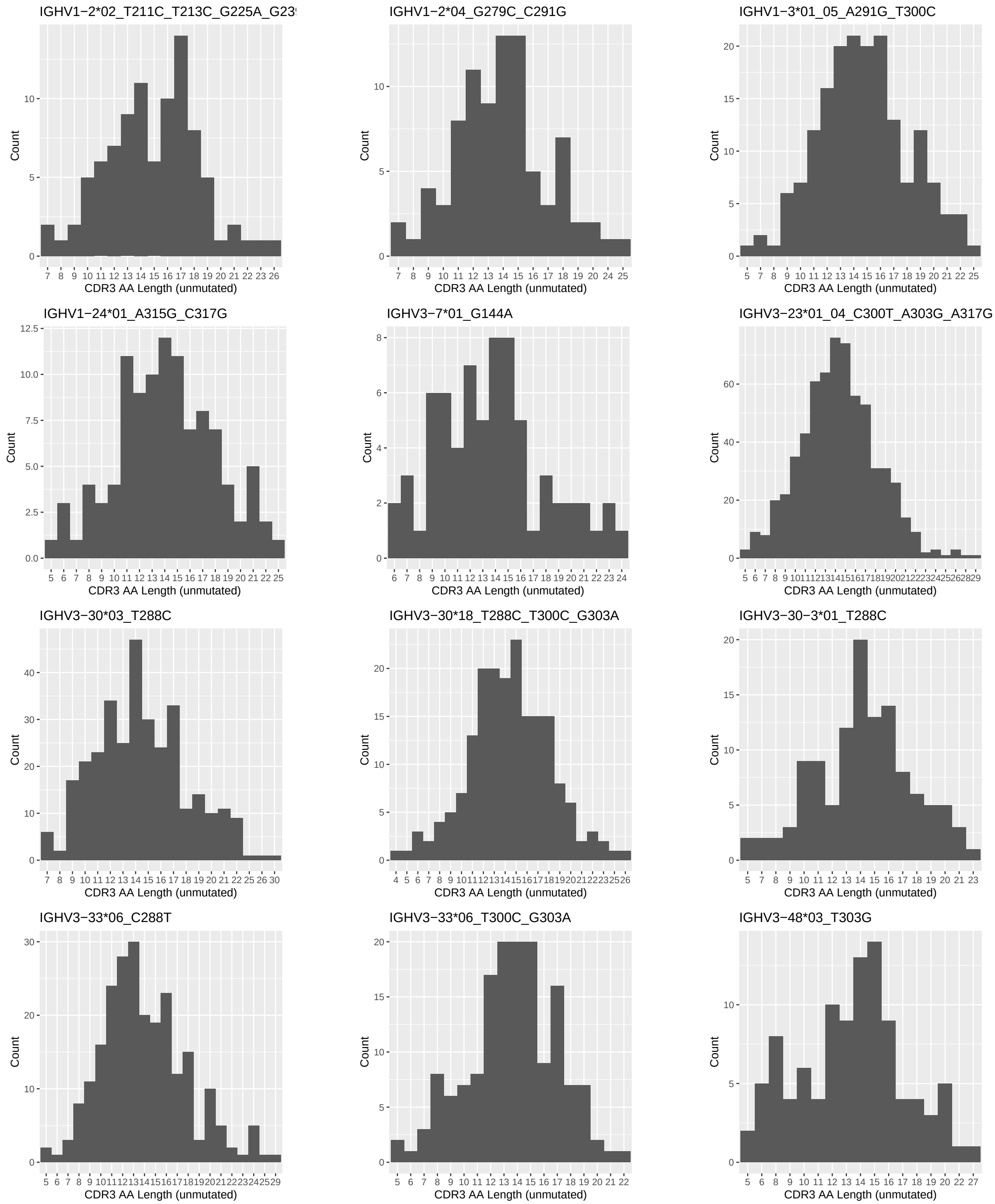


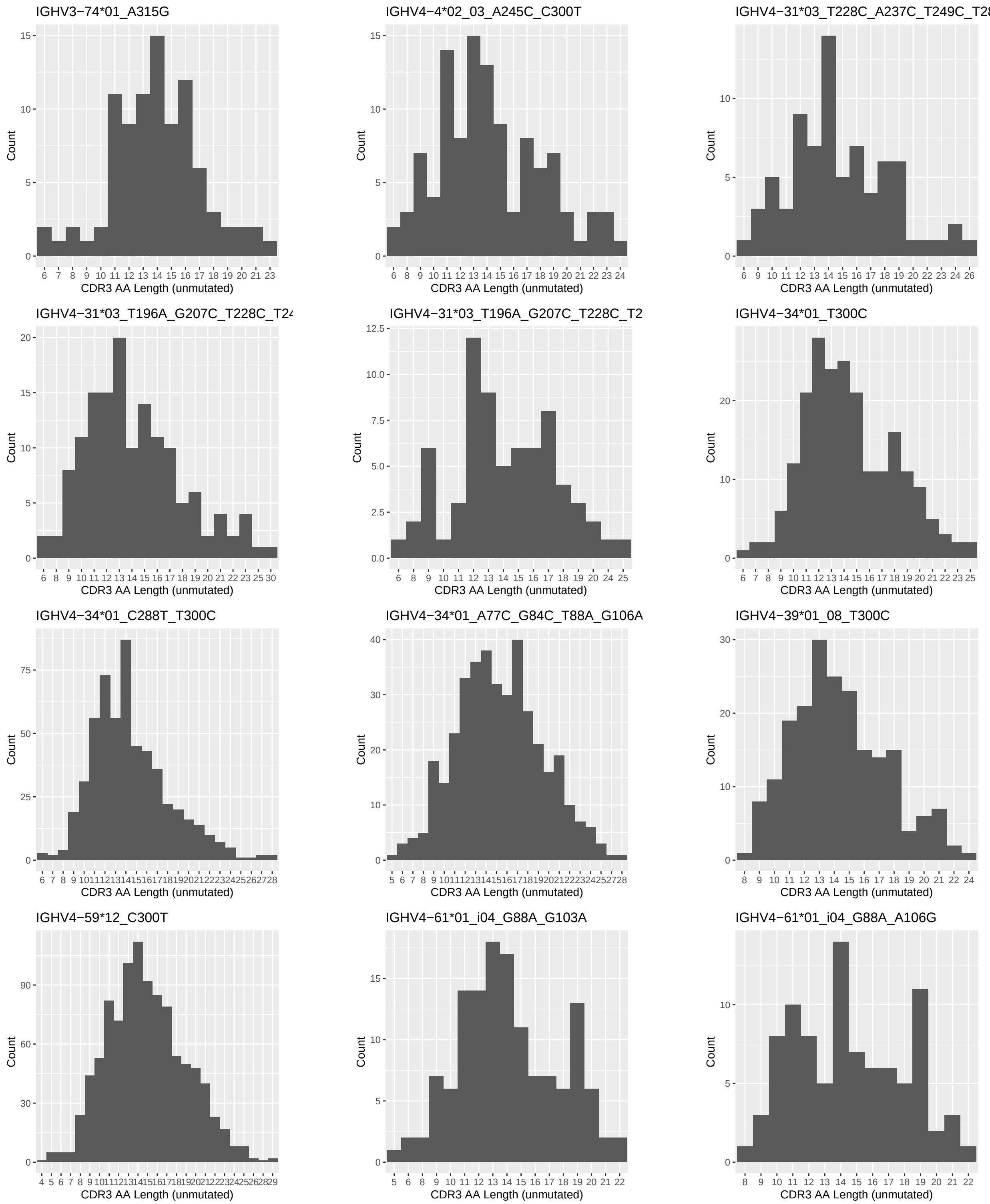
1.4 Final three nucleotides: frequency of each observed triplet

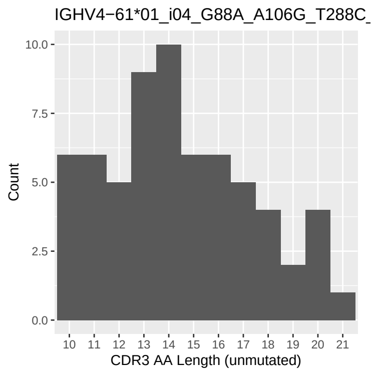


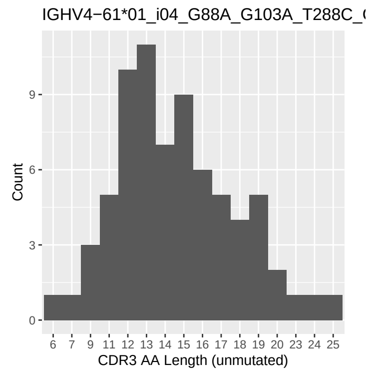

IGHV3−74*01_A315G− Final 3 nucleotides as a triplet
IGHV4−4*02_03_A245C_C300T− Final 3 nucleotides as a triplet
IGHV4−31*03_T228C_A237C_T249C_T285C_G291A_C300T− Fina
IGHV4−31*03_T196A_G207C_T228C_T249C_T285C_C288T− Fina
IGHV4−31*03_T196A_G207C_T228C_T249C_T285C_G291A− Fina
IGHV4−34*01_T300C− Final 3 nucleotides as a triplet
IGHV4−34*01_C288T_T300C− Final 3 nucleotides as a triplet
IGHV4−34*01_A77C_G84C_T88A_G106A− Final 3 nucleotides as a
IGHV4−39*01_08_T300C− Final 3 nucleotides as a triplet
IGHV4−59*12_C300T− Final 3 nucleotides as a triplet
IGHV4−61*01_i04_G88A_G103A− Final 3 nucleotides as a triplet
IGHV4−61*01_i04_G88A_A106G− Final 3 nucleotides as a triplet
IGHV4−61*01_i04_G88A_A106G_T288C_G291A− Final 3 nucleotide
IGHV4−61*01_i04_G88A_G103A_T288C_G291A− Final 3 nucleotide
IGHV4−NL1*01_C300T− Final 3 nucleotides as a triplet
IGHV5−10−1*01_03_A118G− Final 3 nucleotides as a triplet
Count
Triplet

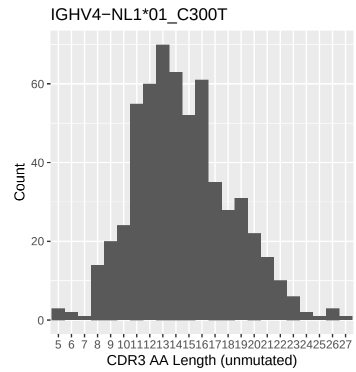
1.5 CDR3 length distribution, in assignments to novel alleles

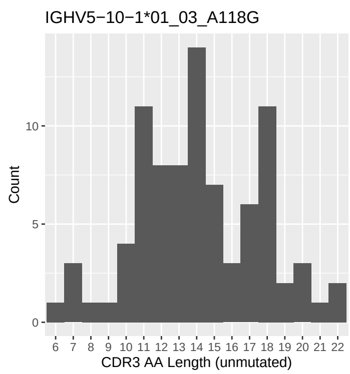



IGHV3−74*01_A315G
IGHV4−4*02_03_A245C_C300T
IGHV4−31*03_T228C_A237C_T249C_T28
IGHV4−31*03_T196A_G207C_T228C_T24
IGHV4−31*03_T196A_G207C_T228C_T2
IGHV4−34*01_T300C
IGHV4−34*01_C288T_T300C
IGHV4−34*01_A77C_G84C_T88A_G106A
IGHV4−39*01_08_T300C
IGHV4−59*12_C300T
IGHV4−61*01_i04_G88A_G103A
IGHV4−61*01_i04_G88A_A106G
Count
CDR3 AA Length (unmutated)

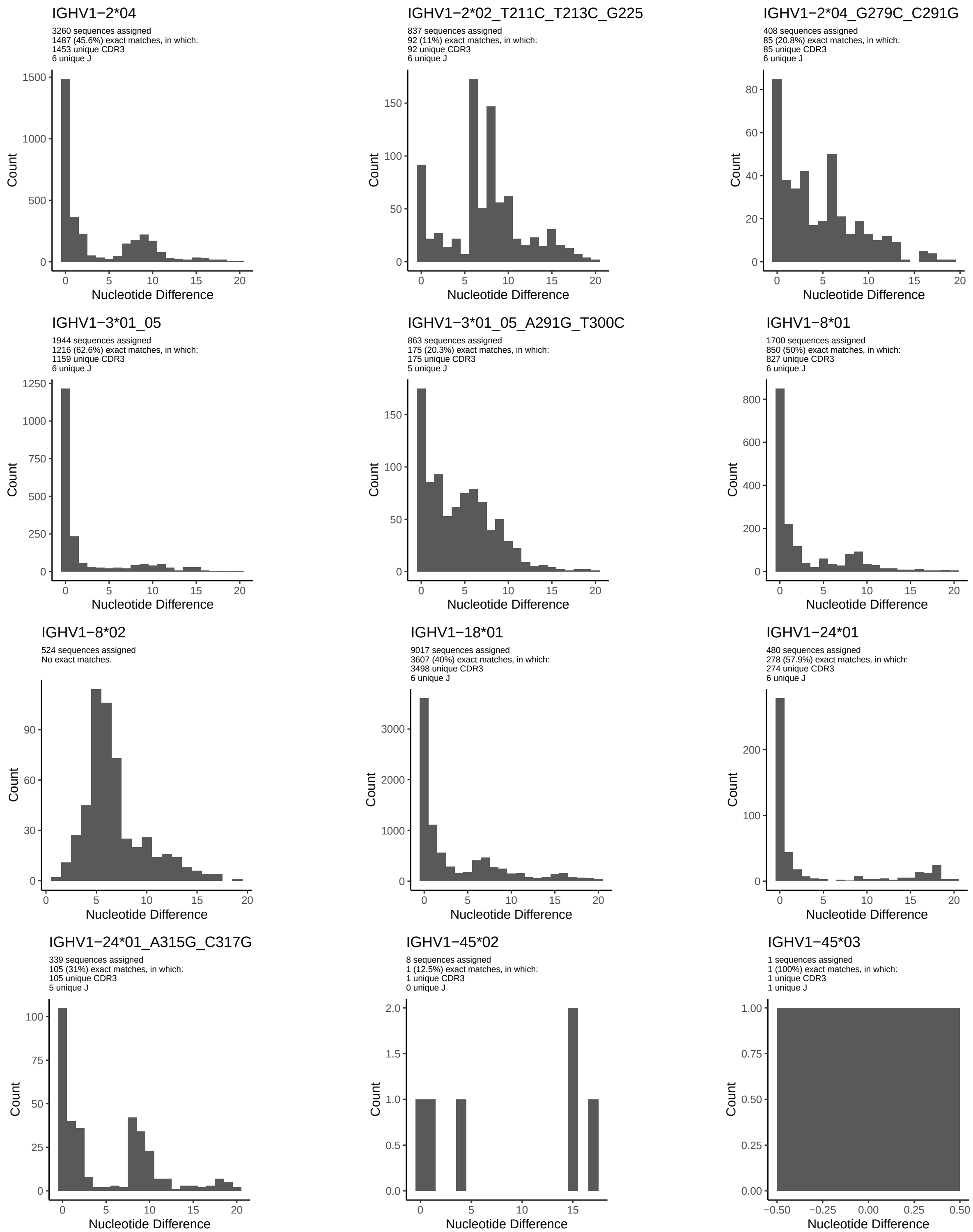

IGHV4−61*01_i04_G88A_A106G_T288C_
Count
CDR3 AA Length (unmutated)

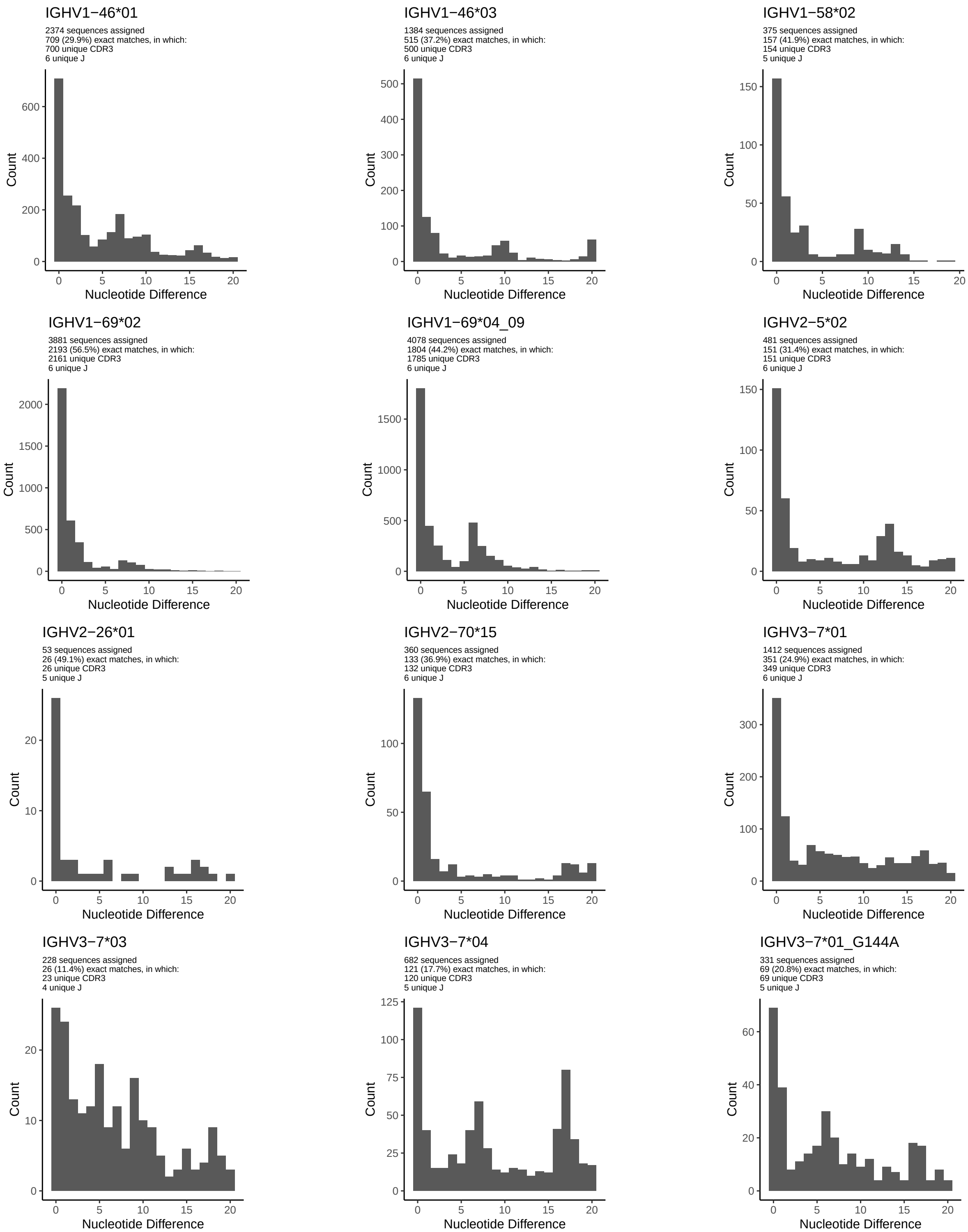

IGHV4−61*01_i04_G88A_G103A_T288C_
Count
CDR3 AA Length (unmutated)

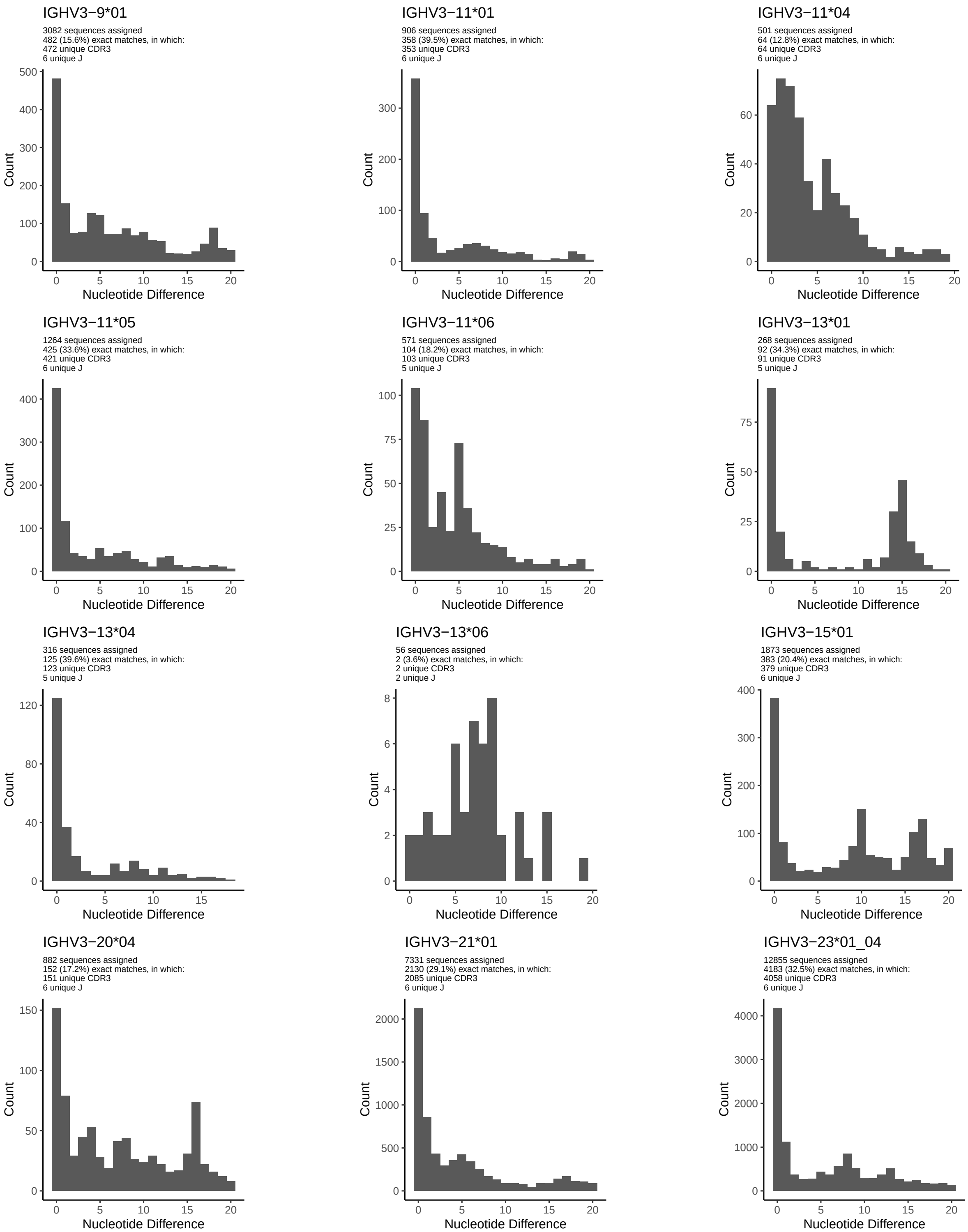

IGHV4−NL1*01_C300T
Count
CDR3 AA Length (unmutated)

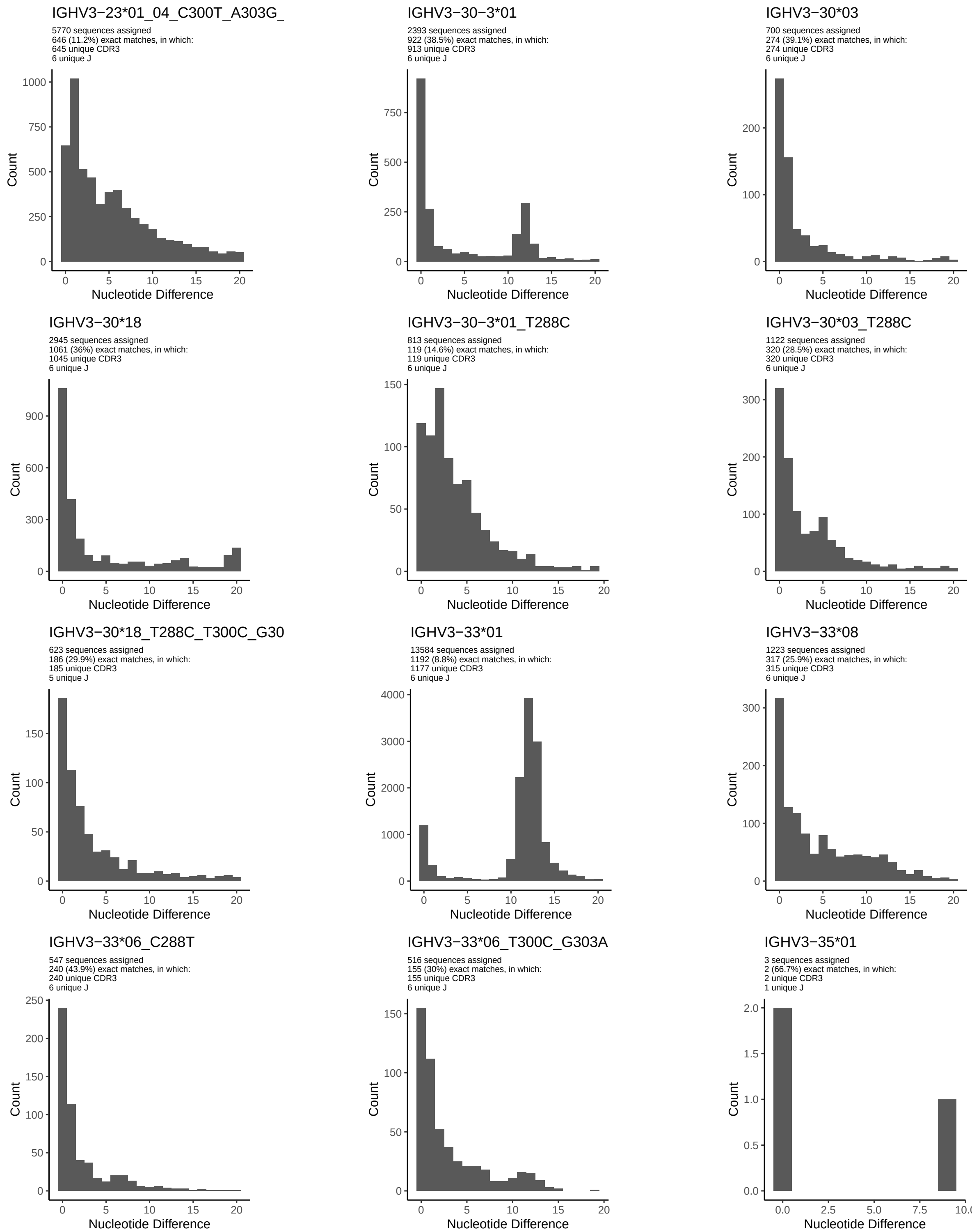

IGHV5−10−1*01_03_A118G
Count
CDR3 AA Length (unmutated)

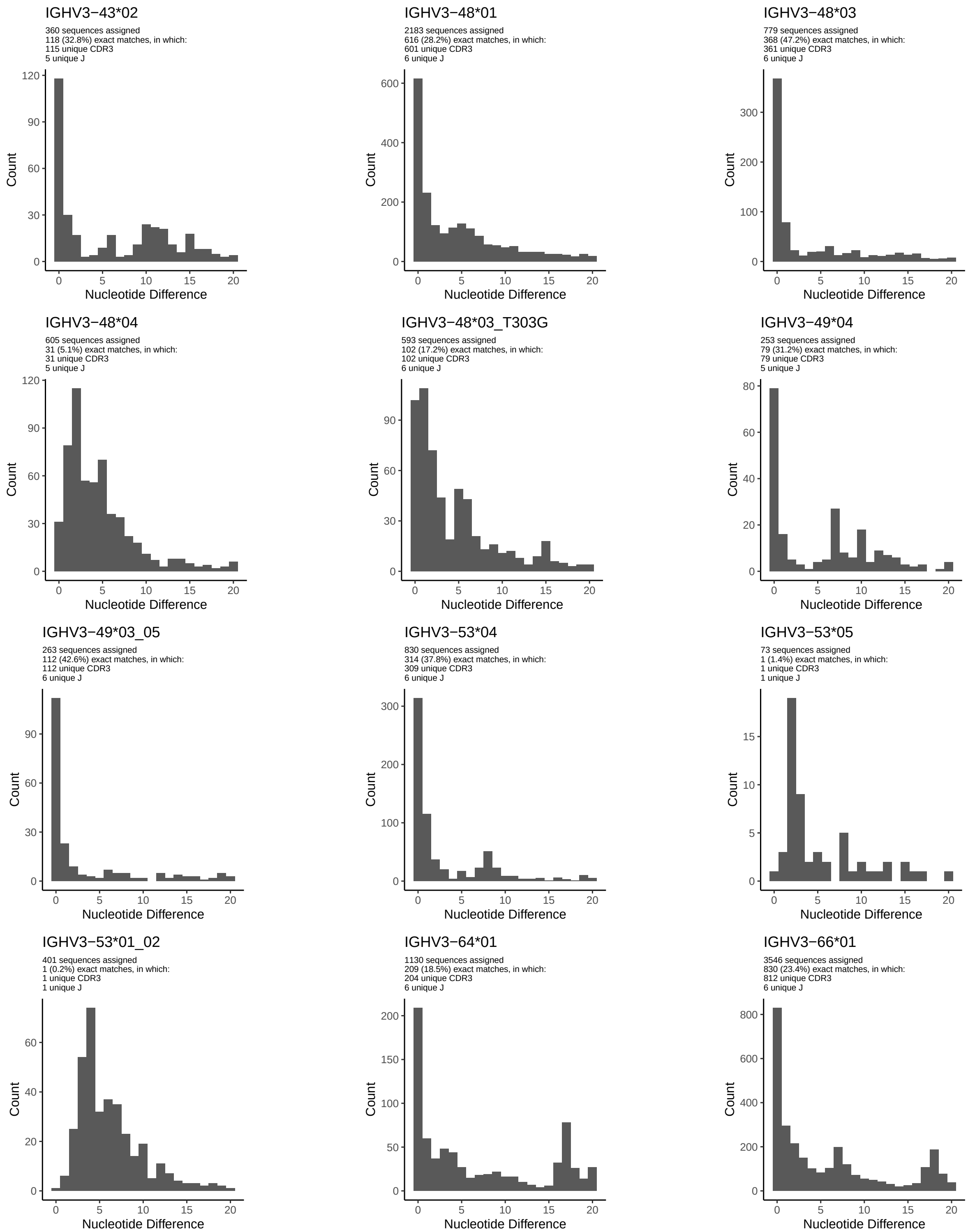
2 Variation from germline, in assignments to each allele

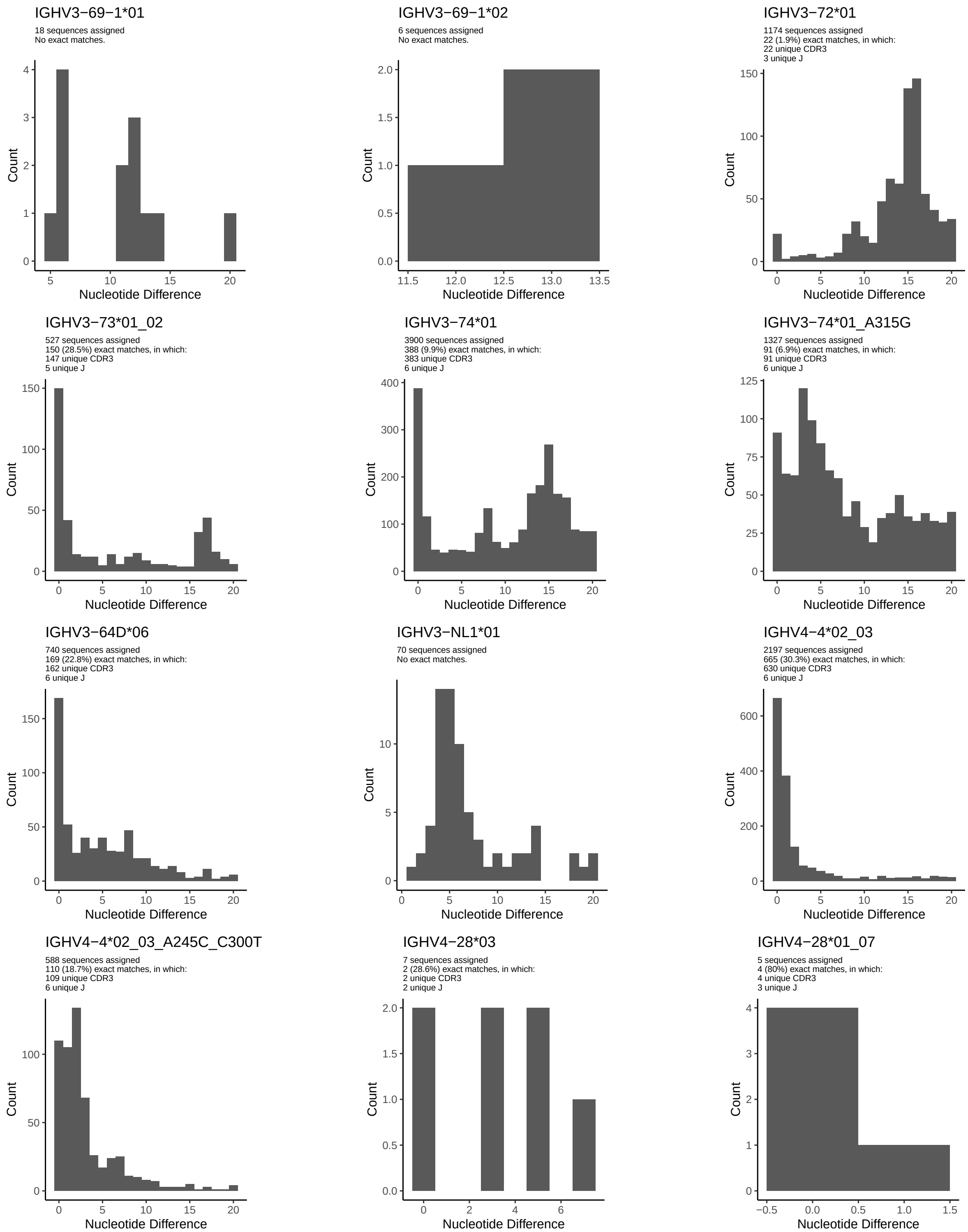


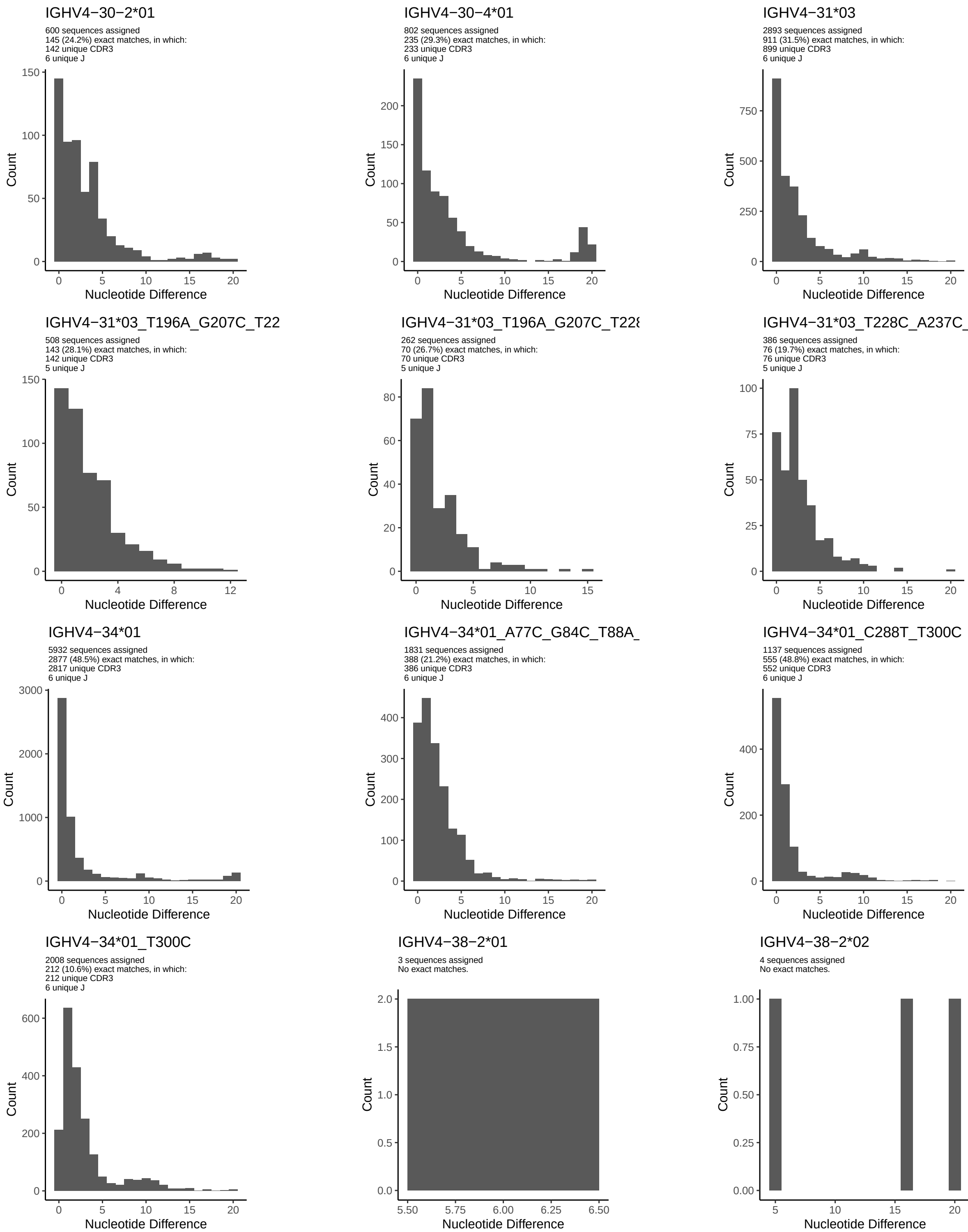

IGHV1−46*01
2374 sequences assigned
709 (29.9%) exact matches, in which:
700 unique CDR3
6 unique J
IGHV1−46*03
1384 sequences assigned
515 (37.2%) exact matches, in which:
500 unique CDR3
6 unique J
IGHV1−58*02
375 sequences assigned
157 (41.9%) exact matches, in which:
154 unique CDR3
5 unique J
IGHV1−69*02
3881 sequences assigned
2193 (56.5%) exact matches, in which:
2161 unique CDR3
6 unique J
IGHV1−69*04_09
4078 sequences assigned
1804 (44.2%) exact matches, in which:
1785 unique CDR3
6 unique J
IGHV2−5*02
481 sequences assigned
151 (31.4%) exact matches, in which:
151 unique CDR3
6 unique J
IGHV2−26*01
53 sequences assigned
26 (49.1%) exact matches, in which:
26 unique CDR3
5 unique J
IGHV2−70*15
360 sequences assigned
133 (36.9%) exact matches, in which:
132 unique CDR3
6 unique J
IGHV3−7*01
1412 sequences assigned
351 (24.9%) exact matches, in which:
349 unique CDR3
6 unique J
IGHV3−7*03
228 sequences assigned
26 (11.4%) exact matches, in which:
23 unique CDR3
4 unique J
IGHV3−7*04
682 sequences assigned
121 (17.7%) exact matches, in which:
120 unique CDR3
5 unique J
IGHV3−7*01_G144A
331 sequences assigned
69 (20.8%) exact matches, in which:
69 unique CDR3
5 unique J
Count
Nucleotide Difference

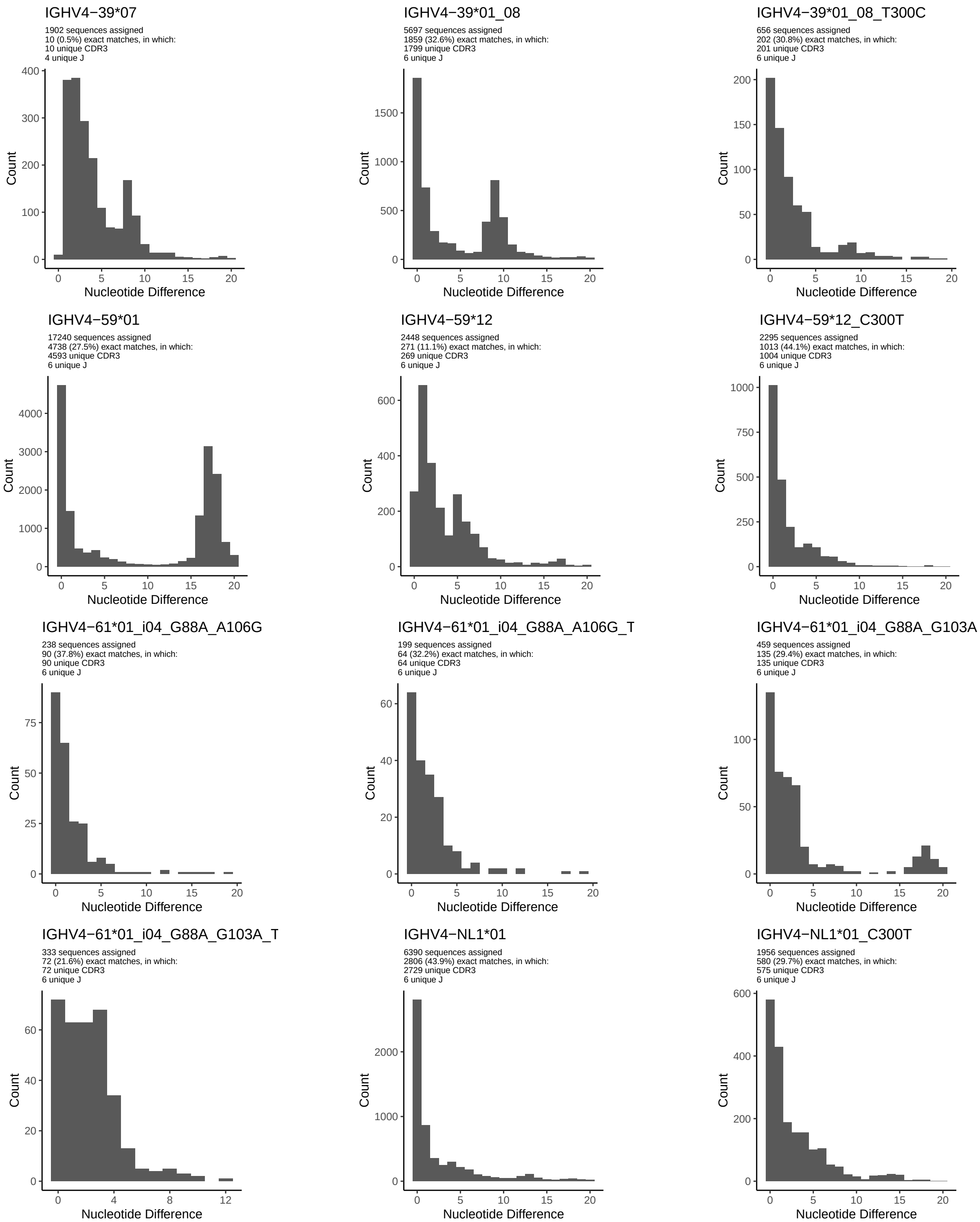

IGHV3−9*01
3082 sequences assigned
482 (15.6%) exact matches, in which:
472 unique CDR3
6 unique J
IGHV3−11*01
906 sequences assigned
358 (39.5%) exact matches, in which:
353 unique CDR3
6 unique J
IGHV3−11*04
501 sequences assigned
64 (12.8%) exact matches, in which:
64 unique CDR3
6 unique J
IGHV3−11*05
1264 sequences assigned
425 (33.6%) exact matches, in which:
421 unique CDR3
6 unique J
IGHV3−11*06
571 sequences assigned
104 (18.2%) exact matches, in which:
103 unique CDR3
5 unique J
IGHV3−13*01
268 sequences assigned
92 (34.3%) exact matches, in which:
91 unique CDR3
5 unique J
IGHV3−13*04
316 sequences assigned
125 (39.6%) exact matches, in which:
123 unique CDR3
5 unique J
IGHV3−13*06
56 sequences assigned
2 (3.6%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV3−15*01
1873 sequences assigned
383 (20.4%) exact matches, in which:
379 unique CDR3
6 unique J
IGHV3−20*04
882 sequences assigned
152 (17.2%) exact matches, in which:
151 unique CDR3
6 unique J
IGHV3−21*01
7331 sequences assigned
2130 (29.1%) exact matches, in which:
2085 unique CDR3
6 unique J
IGHV3−23*01_04
12855 sequences assigned
4183 (32.5%) exact matches, in which:
4058 unique CDR3
6 unique J
Count
Nucleotide Difference

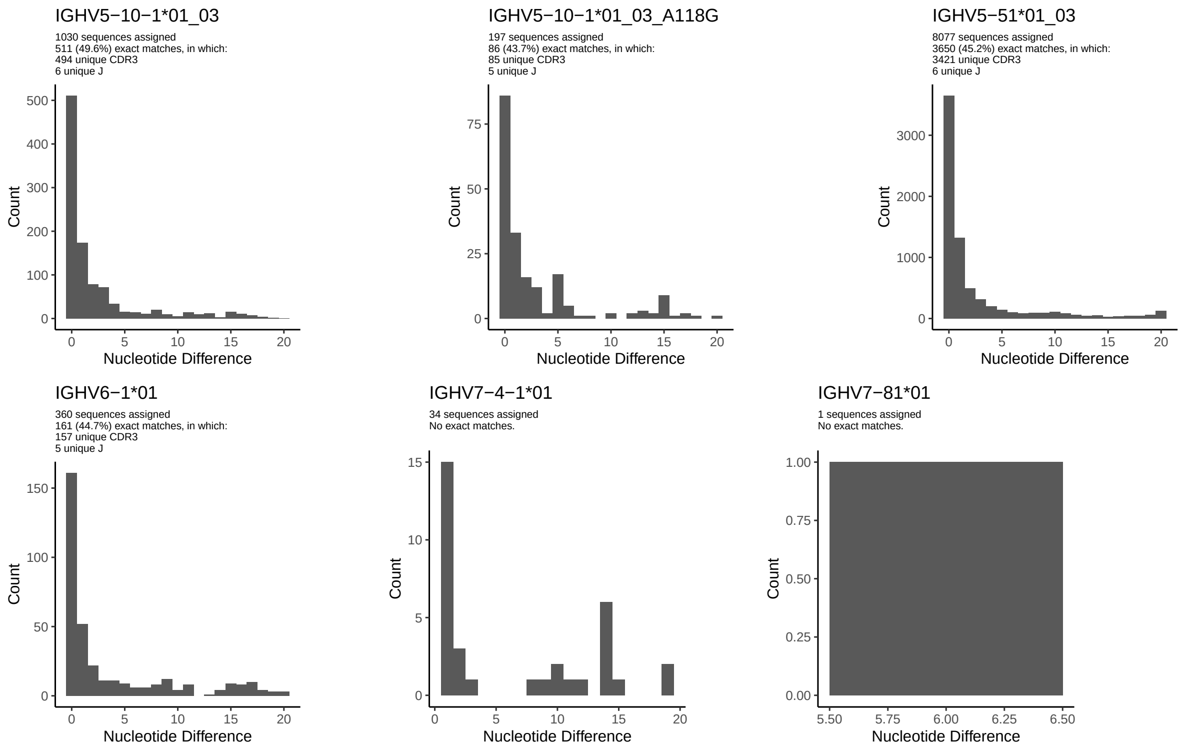

IGHV3−23*01_04_C300T_A303G_
5770 sequences assigned
646 (11.2%) exact matches, in which:
645 unique CDR3
6 unique J
IGHV3−30−3*01
2393 sequences assigned
922 (38.5%) exact matches, in which:
913 unique CDR3
6 unique J
IGHV3−30*03
700 sequences assigned
274 (39.1%) exact matches, in which:
274 unique CDR3
6 unique J
IGHV3−30*18
2945 sequences assigned
1061 (36%) exact matches, in which:
1045 unique CDR3
6 unique J
IGHV3−30−3*01_T288C
813 sequences assigned
119 (14.6%) exact matches, in which:
119 unique CDR3
6 unique J
IGHV3−30*03_T288C
1122 sequences assigned
320 (28.5%) exact matches, in which:
320 unique CDR3
6 unique J
IGHV3−30*18_T288C_T300C_G30
623 sequences assigned
186 (29.9%) exact matches, in which:
185 unique CDR3
5 unique J
IGHV3−33*01
13584 sequences assigned
1192 (8.8%) exact matches, in which:
1177 unique CDR3
6 unique J
IGHV3−33*08
1223 sequences assigned
317 (25.9%) exact matches, in which:
315 unique CDR3
6 unique J
IGHV3−33*06_C288T
547 sequences assigned
240 (43.9%) exact matches, in which:
240 unique CDR3
6 unique J
IGHV3−33*06_T300C_G303A
516 sequences assigned
155 (30%) exact matches, in which:
155 unique CDR3
6 unique J
IGHV3−35*01
3 sequences assigned
2 (66.7%) exact matches, in which:
2 unique CDR3
1 unique J
Count
Nucleotide Difference


IGHV3−43*02
360 sequences assigned
118 (32.8%) exact matches, in which:
115 unique CDR3
5 unique J
IGHV3−48*01
2183 sequences assigned
616 (28.2%) exact matches, in which:
601 unique CDR3
6 unique J
IGHV3−48*03
779 sequences assigned
368 (47.2%) exact matches, in which:
361 unique CDR3
6 unique J
IGHV3−48*04
605 sequences assigned
31 (5.1%) exact matches, in which:
31 unique CDR3
5 unique J
IGHV3−48*03_T303G
593 sequences assigned
102 (17.2%) exact matches, in which:
102 unique CDR3
6 unique J
IGHV3−49*04
253 sequences assigned
79 (31.2%) exact matches, in which:
79 unique CDR3
5 unique J
IGHV3−49*03_05
263 sequences assigned
112 (42.6%) exact matches, in which:
112 unique CDR3
6 unique J
IGHV3−53*04
830 sequences assigned
314 (37.8%) exact matches, in which:
309 unique CDR3
6 unique J
IGHV3−53*05
73 sequences assigned
1 (1.4%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−53*01_02
401 sequences assigned
1 (0.2%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−64*01
1130 sequences assigned
209 (18.5%) exact matches, in which:
204 unique CDR3
6 unique J
IGHV3−66*01
3546 sequences assigned
830 (23.4%) exact matches, in which:
812 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV3−69−1*01
18 sequences assigned
No exact matches.
IGHV3−69−1*02
6 sequences assigned
No exact matches.
IGHV3−72*01
1174 sequences assigned
22 (1.9%) exact matches, in which:
22 unique CDR3
3 unique J
IGHV3−73*01_02
527 sequences assigned
150 (28.5%) exact matches, in which:
147 unique CDR3
5 unique J
IGHV3−74*01
3900 sequences assigned
388 (9.9%) exact matches, in which:
383 unique CDR3
6 unique J
IGHV3−74*01_A315G
1327 sequences assigned
91 (6.9%) exact matches, in which:
91 unique CDR3
6 unique J
IGHV3−64D*06
740 sequences assigned
169 (22.8%) exact matches, in which:
162 unique CDR3
6 unique J
IGHV3−NL1*01
70 sequences assigned
No exact matches.
IGHV4−4*02_03
2197 sequences assigned
665 (30.3%) exact matches, in which:
630 unique CDR3
6 unique J
IGHV4−4*02_03_A245C_C300T
588 sequences assigned
110 (18.7%) exact matches, in which:
109 unique CDR3
6 unique J
IGHV4−28*03
7 sequences assigned
2 (28.6%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV4−28*01_07
5 sequences assigned
4 (80%) exact matches, in which:
4 unique CDR3
3 unique J
Count
Nucleotide Difference

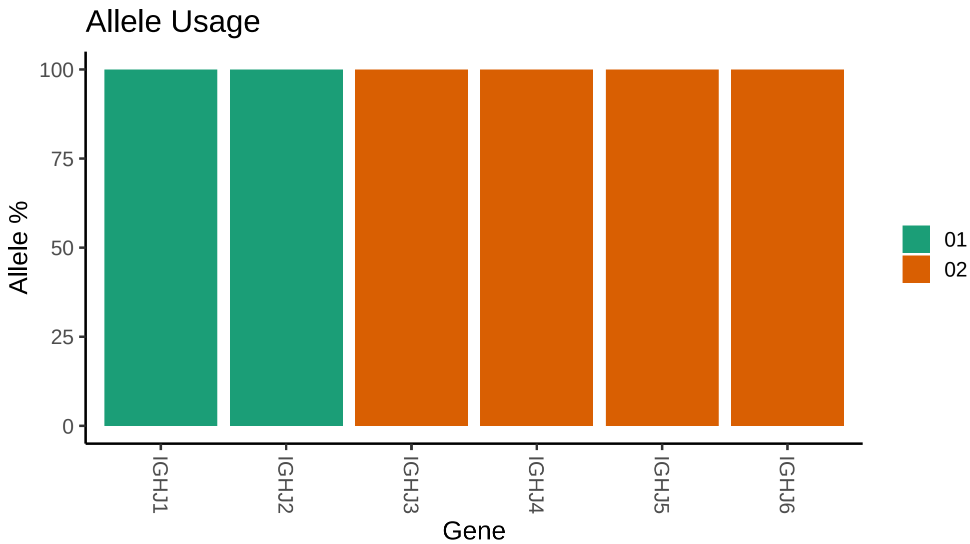

IGHV4−30−2*01
600 sequences assigned
145 (24.2%) exact matches, in which:
142 unique CDR3
6 unique J
IGHV4−30−4*01
802 sequences assigned
235 (29.3%) exact matches, in which:
233 unique CDR3
6 unique J
IGHV4−31*03
2893 sequences assigned
911 (31.5%) exact matches, in which:
899 unique CDR3
6 unique J
IGHV4−31*03_T196A_G207C_T22
508 sequences assigned
143 (28.1%) exact matches, in which:
142 unique CDR3
5 unique J
IGHV4−31*03_T196A_G207C_T22
262 sequences assigned
70 (26.7%) exact matches, in which:
70 unique CDR3
5 unique J
IGHV4−31*03_T228C_A237C_T24
386 sequences assigned
76 (19.7%) exact matches, in which:
76 unique CDR3
5 unique J
IGHV4−34*01
5932 sequences assigned
2877 (48.5%) exact matches, in which:
2817 unique CDR3
6 unique J
IGHV4−34*01_A77C_G84C_T88A_
1831 sequences assigned
388 (21.2%) exact matches, in which:
386 unique CDR3
6 unique J
IGHV4−34*01_C288T_T300C
1137 sequences assigned
555 (48.8%) exact matches, in which:
552 unique CDR3
6 unique J
IGHV4−34*01_T300C
2008 sequences assigned
212 (10.6%) exact matches, in which:
212 unique CDR3
6 unique J
IGHV4−38−2*01
3 sequences assigned
No exact matches.
IGHV4−38−2*02
4 sequences assigned
No exact matches.
Count
Nucleotide Difference


IGHV4−39*07
1902 sequences assigned
10 (0.5%) exact matches, in which:
10 unique CDR3
4 unique J
IGHV4−39*01_08
5697 sequences assigned
1859 (32.6%) exact matches, in which:
1799 unique CDR3
6 unique J
IGHV4−39*01_08_T300C
656 sequences assigned
202 (30.8%) exact matches, in which:
201 unique CDR3
6 unique J
IGHV4−59*01
17240 sequences assigned
4738 (27.5%) exact matches, in which:
4593 unique CDR3
6 unique J
IGHV4−59*12
2448 sequences assigned
271 (11.1%) exact matches, in which:
269 unique CDR3
6 unique J
IGHV4−59*12_C300T
2295 sequences assigned
1013 (44.1%) exact matches, in which:
1004 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_A106G
238 sequences assigned
90 (37.8%) exact matches, in which:
90 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_A106G_T
199 sequences assigned
64 (32.2%) exact matches, in which:
64 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_G103A
459 sequences assigned
135 (29.4%) exact matches, in which:
135 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_G103A_T
333 sequences assigned
72 (21.6%) exact matches, in which:
72 unique CDR3
6 unique J
IGHV4−NL1*01
6390 sequences assigned
2806 (43.9%) exact matches, in which:
2729 unique CDR3
6 unique J
IGHV4−NL1*01_C300T
1956 sequences assigned
580 (29.7%) exact matches, in which:
575 unique CDR3
6 unique J
Count
Nucleotide Difference



IGHV5−10−1*01_03
1030 sequences assigned
511 (49.6%) exact matches, in which:
494 unique CDR3
6 unique J
IGHV5−10−1*01_03_A118G
197 sequences assigned
86 (43.7%) exact matches, in which:
85 unique CDR3
5 unique J
IGHV5−51*01_03
8077 sequences assigned
3650 (45.2%) exact matches, in which:
3421 unique CDR3
6 unique J
IGHV6−1*01
360 sequences assigned
161 (44.7%) exact matches, in which:
157 unique CDR3
5 unique J
IGHV7−4−1*01
34 sequences assigned
No exact matches.
IGHV7−81*01
1 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /work/jenkins/PRJNA491287/M64 045/M64 045/M64 045/M64 045/0f/bc1f3e0640e4eae1d07b0f
##
## Germline reference file: /work/jenkins/PRJNA491287/M64 045/M64 045/M64 045/M64 045/0f/bc1f3e0640e4ea
##
## Novel allele file: /work/jenkins/PRJNA491287/M64 045/M64 045/M64 045/M64 045/0f/bc1f3e0640e4eae1d07b
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 2 04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 2 02_T211C_T213C_G225A_G239C_G240A_G246A_T251C_C252G_C276G_G279
##
##
## Unexpected N in novel sequence IGHV1 2 04_G279C_C291G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 7 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 7 01_G144A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 18: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 30 18_T288C_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_C288T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 48 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 03_T303G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 31 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 31 03_T228C_A237C_T249C_T285C_G291A_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 31 03_T196A_G207C_T228C_T249C_T285C_C288T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 31 03_T196A_G207C_T228C_T249C_T285C_G291A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV4 34 01_C288T_T300C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_A77C_G84C_T88A_G106A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 39 01_08: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 39 01_08_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_G103A: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_A106G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_A106G_T288C_G291A: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_G103A_T288C_G291A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV5 10 1 01_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV5 10 1 01_03_A118G: replacing with gap
```